Supplementary Information for *What spontaneous replicators pass on*

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This supplement contains proofs, generated results and frozen experimental contracts. Paths beginning `results/`, `algocell_exp/` or `manuscript/` resolve from `experiments/`. Source brackets identify the underlying records. Data and code availability and the author's declarations are described in `manuscript/MAIN_v8.md`; local paths and hashes are not public accessions.

Confinement concerns instruction-stream fetches, including operands. Closure concerns dependence on the partner's initial contents under a specified reset, budget and partner class. Neither is a synonym for a loop, copying fidelity, marker carriage or survival. The emulator versions are historical compatibility core C0, the corrected Z80 core (core C1) and the vendor-specific Zilog-NMOS Z80 core (core C2). The Z80 displacement study (registered experiment C1) and the 6502 retention study (registered experiment C2) are separate studies. A tape is a byte sequence; a cell is a population site containing a tape; a genotype is a distinct tape sequence. An update comprises a population encounter batch and its mutation attempts. An execution step is a charged instruction or repeated block operation within an encounter. A sample is a recorded observation. The parent tape is A and the partner tape is B. Worlds are independent evolutionary replicates. Cells, sampling roles, genotypes, substitutions, partners and serial generations are nested observations, not additional worlds.

Supplementary Note 1. Overlap copying fixes a conditional payload count

Ordered overlap theorem and proof

Let a ring contain P byte addresses, with all subscripts interpreted modulo P. Fix a start s and displacement d with 0 < d < P. Starting from memory M_0, perform exactly the ordered assignments

$$M[s+d+k] \leftarrow M[s+k], \qquad k = 0, \ldots, K-1,$$

with no interfering write or read transformation. The byte written at assignment k has original ancestry s+(k mod d). If d divides P and K $\geq$ P$-$d, the complete final ring is the d-periodic tiling of the original interval s,…,s+d$-$1.

Proof. For k < d, no earlier write has reached source s+k: its first possible write is assignment P$-$d+k, which exceeds k. It therefore still contains its original byte. For k $\geq$ d, the most recent write to source s+k occurred at assignment k$-$d. Writes to that physical address are separated by P assignments; because d < P, no later congruent write precedes k. By induction the value at that source is the original byte at s+((k$-$d) mod d) = s+(k mod d). This proves the ancestry statement for every write, including after wrapping. If d divides P, a destination with physical offset j = d+k mod P has the same residue modulo d as k. Every write therefore supplies the original byte of its own residue class. The first P$-$d writes cover offsets d,…,P$-$1; the first d positions retain their own original values or subsequently receive copies of those same values. Later writes preserve the tiling. Thus every address has the asserted value and ancestry. ■

For two length-L tapes, take P = 2L. When d divides L, the physical partner position q has ancestor s+((L+q−s) mod d), provided the protected root interval lies in the original parent. With s = 0 this is q mod d. Let C be the independently established constrained coordinates within a protected d-byte seed; all other seed bytes must be jointly arbitrary without changing execution, coverage, offspring viability or entry machinery. The free coordinate set is F = {0,…,d−1}\C, with r = d−|C|. Each assignment to F produces a distinct offspring because every seed coordinate occurs in the covered output. There are therefore 256^r members of this product family, or $\log_2(256^r) = 8r$ ideal noiseless bits. This is a conditional mathematical count, not a measured Shannon capacity, a claim of neutral fitness or an exhaustive intervention result. Independent validation of family membership after reset permits the same argument in every generation. [Proof specialization: Supplementary Note 1, ordered overlap theorem and Z80 specialization; generated geometry: `results/capacity/geometry_traces.csv.gz`]

Z80 specialization of the ordered overlap theorem

This specialization uses tape length $L \leq 64$, a 2L-byte encounter ring, zero initial registers and flags except the specified stack pointer, physical entry zero and a budget of 128 execution steps; each `LDIR` repetition consumes one step. The parent begins with the four bytes `XX 5e ed b0`. Set d = XX mod 2L and require $4 \leq d \leq L$ and d dividing L. The initial one-byte instruction XX must pass control to address 1, must preserve every byte below d and must leave $HL \equiv 0$ and $256D \equiv 0$ modulo 2L, with BC either zero or at least 126. Architectural source and destination increments must preserve the specified constant displacement in the mapped ring throughout the remaining copies. This last requirement excludes an unexamined architectural wrap that changes the mapped displacement; congruence of the initial pointers alone is insufficient.

Under these conditions the whole encounter leaves both tapes equal to the d-periodic tiling of the parent's original first d bytes. Every assignment to coordinates 4,…,d−1 is transmitted jointly and serially, and initial coordinates d,…,L−1 are overwritten before contributing ancestry to the offspring. Exactly d−4 coordinates outside the held four-byte core are consequently free.

Proof. The first instruction preserves the protected prefix and hands control to `LD E,(HL)` at address 1. That instruction reads the original byte XX at mapped address zero. Since $256D \equiv 0$ modulo 2L, DE now maps to d, while HL maps to zero. There are 126 executions left. BC = 0 means a decrementing block operation has 65,536 iterations before exhaustion; $BC \geq 126$ likewise permits all remaining iterations. With the architectural-increment requirement, iteration k reads physical k mod 2L and writes (d+k) mod 2L. The ordered overlap theorem (Supplementary Note 1) gives ancestry k mod d. Because d divides 2L, writes to core addresses 0,…,3 restore their own original bytes, so the still-executing block instruction is preserved. The 126 copies cover the ring because $2L-d \leq 128-4 = 124 \leq 126$. The first instruction's possible writes outside the prefix disappear before they are read by the moving copy. Finally d dividing L aligns B with the parent seed at physical entry zero. Free seed bytes are data, and the fixed execution certificate makes their simultaneous values irrelevant to control; descendants inherit both those bytes and the same core. Reset therefore puts each descendant in the same admissible family, proving serial inheritance by induction. Initial coordinates at or above d are overwritten from the seed before being read, establishing erasure of their original ancestry. ■

The proof checks that admitted one-byte setup instructions under this reset do not introduce a harmful 16-bit pointer wrap during the block operation. The candidate and setting ledgers also identify the executed code and geometry. The immediate form `1e NN ed b0` has d = NN mod 2L

and 127 copying executions after `LD E,n`; it has the same four held core bytes and payload formula under the corresponding hypotheses. Reverse `LDDR` does not inherit this forward proof. [Current certificates: `results/capacity/candidate_eligibility.csv`, `results/capacity/edit_eligibility.csv`; derivation: Supplementary Note 1, ordered overlap theorem and Z80 specialization]

The conditions are substantive. At L = 16, `0x70` (`LD (HL),B`) nominally gives d = 16 and reaches address 1, but overwrites source byte zero with zero, so `LD E,(HL)` loads zero and the claimed copy fails. Taken returns such as `RET NZ` can preserve registers while violating the entry condition and using partner-dependent stack contents. With $d \leq 3$, early overlap can destroy the block instruction. If d divides 2L but not L, the ring can tile correctly while the physical offspring begins in a different phase, so serial entry and the $d-4$ result do not follow. If d does not divide 2L, wrapped writes can place different residue classes onto the core. If coverage is insufficient, original partner bytes can survive. The stated family excludes these cases before retention is assessed.

Holding the four core coordinates fixed does not establish that they cannot transmit alternative alleles. Some core substitutions preserve a working mechanism, including alternative offset-setting opcodes. A thresholded single-mutant site score can include such machinery alleles or lose genuine copied data under its whole-tape qualification gate; it is therefore distinct from the fixed-footprint payload count. Visible repetition of a baseline tape likewise need not equal its effective copying displacement. [Generated comparisons: `results/crystal/cell_exceptions.csv`, `results/crystal/offset_examples.csv`; `results/offset/offset_cells.csv`]

Supplementary Note 2. Provenance certifies ancestry and serial closure

A sufficient family certificate

A certificate specifies a family of parent tapes, the held code/dependency coordinates, reset state, allowed partner inputs, instruction budget, encounter termination, physical offspring boundary and entry phase. Initially every parent and partner byte receives a distinct symbolic origin. A pure copy inherits its source's origin at the time of the read. If the source byte was overwritten, the copy inherits the origin carried by that write. Matching numerical byte values never merge original coordinates. Register transfers preserve labels only where the semantics prove an exact byte transfer. Unknown transformations and arithmetic on values cannot be silently treated as literal copying.

For every jointly admitted payload assignment and partner, the certificate must establish that the ordered execution remains valid, every physical offspring byte has the predicted parent source, no original-partner input reaches the relevant output and the copied machinery and physical entry remain viable. It must also prove that reset places that actual offspring within the same certified family. A single baseline trace supports extraction of a candidate certificate; it cannot by itself prove noninterference for arbitrary values. An unmodelled instruction or dependency remains unresolved, rather than being inferred away from observed retention. [Certificate implementations and ledgers: `results/capacity/candidate_eligibility.csv`; `results/evolved6502/eligibility.csv`; `results/s8086/prelaunch/ASTRA_PRELAUNCH_REVIEW.md`]

Serial certificate theorem. Suppose every allowed parent/partner encounter in a family has the same certified offspring source map, independent of the partner's unknown contents, and every offspring belongs to that family after reset. Then its ancestry follows successive composition of that map. To conclude preservation of free information, additionally require that the induced

offspring map is injective on admitted free assignments and that its output again represents an assignment in the same certified family. The recovery rule must be common to the declared partner class; conditional invertibility requiring knowledge of an unobserved partner is insufficient.

Proof. The first encounter satisfies the certificate by hypothesis. Its actual offspring belongs to the same family, so the certificate applies to the second encounter with its independently allowed partner. Repeating proves the ancestry claim by induction. For information preservation, each per-encounter map is injective on the admitted family, and a finite composition of injective maps remains injective. The starting free assignment therefore remains recoverable after every finite number of generations. In the protected-seed specialization, every free root is represented in the covered descendant and the idempotent tiling map preserves its first-generation canonicalization. The proof requires continued offspring membership in the certified family, noninterference and recoverability. It does not depend on the number of sampled partners. ■

This is a sufficient criterion for a pure-copy family. Arithmetic encodings, changing programs and more general channels may transmit information while falling outside it. Failure to certify them is not proof that they lack heredity.

Shifted source roots in amended S2 L2

The sampled baseline is d-periodic at physical entry zero, with d dividing L. Ordered writes have geometry s+k→s+d+k modulo 2L. The original root interval R = {s,…,s+d−1} must lie within A; full B coverage, no original-partner read and no interfering write are required. For each physical offspring position q, the predicted source is

$$a(q) = s + ((L + q - s) \bmod d).$$

The raw held footprint K is extracted before retention. Its bound c = 1+max(K) ≤ d is retained, but the exact constrained root set is C_R = {a(k):k∈K}. Free roots are F = R\C_R; erased coordinates are E = A\R. The exact count is d−|C_R|, which need not equal d−c. Both K and C_R are held. Mutable erasures are E\K; held-erased coordinates E∩K can govern setup even though their original ancestry disappears. Dependencies must not be reduced modulo d to conceal a physical setup requirement. All comparisons use B's physical coordinates, without a favourable rotation.

[Prospective amendment: `REVISION_PREREG.md`, S2 B2; certificate review: `results/s8086/prelaunch/ASTRA_PRELAUNCH_REVIEW.md`]

The baseline periodicity makes A[k] = A[a(k)] for each held coordinate. With K and C_R fixed and the complete execution certificate intact, copying reconstructs every held byte at physical entry. Each free root's assignment is reproduced at its predicted output coordinates. As d divides L, a(a(q)) = a(q), so the map composes with itself. These facts establish the serial family condition; copying values that happen to match the core in one trace would not establish it.

Inflow and confinement are different properties

Fix the parent's tape X = x and draw a partner E from the declared distribution. Let Y be the full physical partner tape after execution. With deterministic execution conditional on x and E, the information inflow is I(E;Y|X = x) = H(Y|X = x). Perfect copying, P(Y = x|X = x) = 1, implies H(Y|X = x) = 0 and hence zero inflow. The converse fails: a regenerator can produce the same canonical output for every partner while erasing an introduced parental change. These two

statements distinguish zero inflow from faithful inheritance. If execution has additional randomness, entropy alone also includes that randomness and cannot be identified with partner mutual information without the corresponding conditioning.

The empirical estimate is plug-in offspring entropy over 256 partners. Its sample ceiling is $\log_2 256 = 8$ bits; a ceiling value does not establish an eight-bit bound on the true output diversity, and zero over the sample is sampled independence. Confinement does not guarantee zero inflow because unwritten offspring bytes can retain partner contents. Zero inflow does not require confinement if all accessed partner contents have first been overwritten independently of their originals. The BFF literal-bandwidth controls supply this dissociation. [Generated inflow records: `results/biology/individuality/`; `results/inflow_insoup/per_world.csv`; `results/bff/`; `results/bff_dials/`]

Supplementary Note 3. Literal-channel results have restricted scope

Copy cost and repeated instruction addresses

In unmodified BFF, a pair has 128 byte positions; the instruction pointer and both data heads start at zero, and each head movement changes its position by one byte. An arithmetic or copying instruction changes at most one byte and does not also move a head. To make B an exact copy at a fixed shift when every original B byte differs from the desired copy requires at least 64 changing instructions. The heads must collectively visit all 64 B byte positions; from their shared initial position zero, v moves can expose at most v new positions, so at least 64 head moves are also required. The sets of operations are disjoint, giving at least 128 instruction executions. If all executions stay within A's 64 addresses, some address is revisited. Without a bracket transfer the instruction pointer advances; the return requires a taken backward bracket jump. This is a repeated address, not a periodic full machine state.

More generally, an execution confined to L instruction-start addresses repeats an address whenever it completes a copy costing more than L executions, or it cannot terminate inside A before a budget $T > L$ is spent. Otherwise at most L distinct starts could execute, contradicting the premise. In the historical benign Z80 harness a held `HALT` counts as repeated residence at its own address; this convention establishes repetition, not a dynamically traversed loop. Lethal zero-opcode stopping and budgets at or below L invalidate the budget premise. The BFF copy-cost argument does not rule out a straight pass over all 128 pair addresses. Earlier proposed uniform-partner and zero-partner straight-line bounds were withdrawn because partner-dependent tests and writes ahead of the instruction pointer invalidate their proofs.

A self-consistent literal-write instruction or macro can construct a straight-line self-writer when its prescribed stream is executed and its writes cover the partner with the correct phase and budget. Default instruction flow makes such a program open only if execution actually reaches the partner before stopping. A different conditional statement says that if any execution visiting each instruction address at most once cannot complete a copy in the specified context, every self-writer repeats an address. The historical default-flow application also assumes execution ends upon leaving the pair. That copy-cost premise is not established for BFF by the preceding bound, and the exit premise fails for the ring Z80. These constructions do not prove which replicator evolves first.
[Constructions and scope: `results/literal_channel/CHECK.md`]

Repeated load–push pair theorem

Let a load opcode a consume w following literal bytes L, setting one data register R verbatim. Let the distinct one-byte push opcode b emit a fixed permutation π(L), preserve R and move a separate stack pointer by $-$w. Assume the executed stream remains E = (a L b)$^\infty$, consecutive pushes occupy contiguous w-byte blocks and the final compared region consists of those writes without holes, unrelated writes or source-stream modification. For a finite copied window of N bytes require

$$N \geq (w+2) + w - \gcd(w+2, w) = 2w + 2 - \gcd(2, w).$$

Under these hypotheses, odd w cannot produce an exact copied window. Even w admits precisely the compatible alternating layouts: E = (a b)$^\infty$, L = (b a)$^{w/2}$ and π(L) also alternates. For an infinite exact stream the conclusion needs no finite-window bound.

Proof. Each push emits the same block C = π(L). Reading the descending stack in ascending address order reverses the order of successive blocks, not the internal layout of each block, so the output is C$^\infty$ up to phase. A matching N-byte window has periods w+2 and w. The Fine–Wilf periodicity theorem [15] at the stated length gives period g = gcd(w+2,w) = gcd(2,w). The window covers a complete execution period, so the g-periodicity extends to E. For odd w, g = 1 forces a = b, contradicting distinct opcodes. For even w, g = 2 forces alternation; byte zero is a and byte w+1 is b, giving the stated L. The output permutation must preserve alternation. Conversely any such permutation writes a shift of E, proving the stream construction. Actual encounter bootstrap and coverage remain additional conditions. ■

The finite bound is 2w+1 for odd widths and 2w for even widths. The shorter odd-width bound 2w is insufficient: w = 1 permits execution period `a a b` and a two-byte output `a a` before disagreement. At w = 3, execution `a b a a b`, literal `b a a` and permutation (1,0,2) emit block `a b a`; the six-byte window `a b a a b a` matches before disagreement. These are explicit counterexamples to the shorter finite bound. An independent symbolic equality check covers all 46,233 layouts for widths 1,…,8. For w = 2r there are 2(r!)^2 compatible layouts; both width-two permutations work. [Generated records: `results/literal_channel/pair_layouts.csv`, `results/literal_channel/finite_bound_counterexamples.json`, `results/literal_channel/verification.csv`]

The theorem characterizes a restricted literal channel, independently of the default-flow/copy-cost results. Loads to the stack pointer are excluded: their value changes addressing, and the original-8086 `PUSH SP` stores the decremented pointer. Transient pushes using initial register contents cannot be silently substituted for the recurrent load–push stream. The Z80 words `01 c5`, `11 d5` and `21 e5` are width-two immediate-load witnesses. The other two nontrivial period-two census words, `2a e5` and `e5 2a`, use a memory load and are outside this literal grammar. Thus three, not five, of the five nontrivial words are literal-channel witnesses. The validated census and causal classification define the 8086 inventory used here (Supplementary Note 8). [Generated Z80 agreement: `results/literal_channel/census_audit.csv`, `results/literal_channel/census_provenance.json`; current 8086 census: `results/census8086/table_main.csv`, `results/census8086/table_hits.csv`]

Width-one mixtures cannot write the same infinite periodic stream

Allow arbitrary finite sets of distinct load and push opcodes, with disjoint opcode sets. Every load consumes one literal byte and sets the same one-byte register; every push emits that register

unchanged and advances the separate descending pointer by one. Decode a prescribed periodic input sequentially. Exclude the degenerate constant all-push writer supplied with an initial register already equal to its push opcode. There is no nondegenerate infinite periodic self-writer in this model.

Proof. Let d be the input's primitive byte period. At a token boundary, opcode identity determines whether the next boundary is one or two bytes later, independently of R. The finite graph of boundary residues modulo d therefore eventually enters a directed cycle with $t \leq d$ distinct vertices. If that cycle has $n_L > 0$ loads and $n_P > 0$ pushes, then $n_P < t \leq d$. Loads recur identically in each traversal, so the emitted word per cycle has length n_P; reversing its chronological order for the ascending-memory stack view still gives an output of period n_P. If this infinite output equals a shift of the input, n_P must be a positive multiple of the primitive period d, contradicting $0 < n_P < d$. An all-load cycle produces no unbounded output. An all-push cycle visits every residue and emits a constant, matching only the excluded constant opcode word with its specially initialized register. Finite transients cannot alter the recurrent contradiction. Multiple opcode identities do not change the boundary argument. ■

The proof is not a theorem about every finite 6502 encounter, every instruction family or all nonperiodic streams. In particular it neither proves a Toeplitz structure for arbitrary mixtures nor proves that all 6502 replicators are confined. The symbolic minimal-cycle search records 992 width-one count cases through execution period 64, all unsatisfiable without unknowns. The independent union-find enumeration directly checked 165,580,120 token orders through period 40. These searches check the restricted theorem; the 17,167,680,177,532 unreduced orders covered by the reduction were not enumerated individually. [Generated records: `results/literal_channel/minimal_symbolic_search.csv`, `results/literal_channel/direct_search.csv`, `results/literal_channel/direct_run.json`, `results/literal_channel/verification.csv`]

For every $w \geq 2$, mixtures provide a different construction: with identity byte layout, the input $(a\, b^{w-1})^\infty$ executes a load of $b^{w-1}a$ followed by $w-1$ pushes. Its execution cycle consumes 2w bytes, and every output block is a shift of the same period-w word. At w = 3 this is input `(a b b)^∞`, decoded `a [b b a]; b; b`, which emits repeated `b b a`. It refutes an odd-width prohibition for arbitrary mixtures while respecting the repeated-pair theorem's narrower scope. [Constructions: `results/literal_channel/minimal_witnesses.json`, `results/literal_channel/direct_witnesses.json`]

A finite width-one factor witness further separates infinite equality from bootstrap. The 16-byte input `a9 a9 a9 48 48 a9 a9 48 48 48 a9 a9 a9 48 48 48` decodes `LLPLPPPLLPP`. Its five loads and six pushes produce a matching 16-byte factor at a cyclic shift, but disagree at byte 17; the input primitive period is 16 and output period six. The Fine–Wilf threshold [15] for those periods is $16+6-\gcd(16,6) = 20$. In the actual zero-partner 6502 ring this tape fails at budgets 39, 128 and 512 under both BRK rules: the presumed unmodified infinite tiling does not bootstrap when the pointer reaches virgin partner bytes. All six replays are retained. [Generated witness and replays: `results/literal_channel/finite_witness.json`, `results/literal_channel/finite_search.csv`, `results/literal_channel/6502_finite_harness.json`]

Supplementary Note 4. Machine-specific laws and intervention denominators

Constructed 6502 and evolved retention

The six-byte constructed 6502 cores are `b5 zz 48 ca 50 fa` (`LDA zp,X; PHA; DEX; BVC`) and `ba b5 zz 48 50 fa` (`TSX; LDA zp,X; PHA; BVC`). The reset is A = X = Y = 0, P = 0x20 and PC = 0, with the first push landing at physical $2L-1$. Data reads use architectural eight-bit zero-page indexing; pushes use page-one address 0x100|S. Traversal is descending and the census defines $d = (\text{read}-\text{write}) \bmod 2L$, opposite to the forward theorem's destination-minus-source convention. At iteration i, $S_i = (S_0-i) \bmod 256$, with S_0 = 0xFF at L16/L32 and 0xDF at L20. The first core reads $(zz-i) \bmod 256$ and the second reads $(zz+S_i) \bmod 256$; both write $0x100|S_i$ before mapping to the ring. This actual arithmetic, including the separate zero-page and stack wraps, predicts displacement before retention outcomes are examined. It must remain constant throughout the executed copy. The further conditions are $6 \leq d \leq L$, d dividing L, preserved six-byte machinery, the required ordered traversal and complete output coverage without interference. Under these hypotheses the copied prefix is coordinates $0,\ldots,d-1$ and the free payload outside the held core is $d-6$. A d-periodic parent can already equal its regenerated output even when $d < L$; the reported whole-tape exactness comparison concerns the sampled random bodies. [Geometry implementation: `census6502_report.py`, `predicted_offset`]

The constructed census checks every operand at L = 16, 20, 32, both faithful and halt-before-BRK rules, budgets 128/512, three random bodies and 64 partners per body. Its 296 qualifying rows satisfy the sampled aligned-carriage prediction. They are constructed cases with sampled bodies and partners, not the all-value exact capacity interventions of the 6502 retention study. At a 40-byte ring, 256 mod 40 = 16, so eight-bit pointer wrapping can destroy constant displacement; a finite successful copy outside the hypotheses does not extend the law. [Generated geometry and checks: `results/census6502/closers_scan.csv`, `results/census6502/table_closers_law.csv`, `results/census6502/table_closers_exact.csv`; derivation: `results/census6502/CENSUS.md`, section 8]

The 6502 retention study uses the complete corrected S1 panel: 62 first-detection roles, 640 final random-cell draws and 240 final top-three roles, totalling 942 roles from 80 worlds and 861 distinct tapes. Exactly 42 roles qualify, representing 41 distinct tapes from 11 worlds. The footprint counts constrained coordinates, which need not form a contiguous prefix. Ordered pure-copy labels are propagated through admitted loads, stack operations, transfers and stores. Unknown arithmetic, reverse or nonconstant motion, partner input, interference and uncertified control remain excluded or unresolved. Each world uses its frozen machine with the intervention budget fixed at 128, including candidates originating in S1 arm C's 512-instruction worlds. [Generated panel: `results/evolved6502/panel.csv`, `results/evolved6502/eligibility.csv`, `results/evolved6502/geometry.csv`, `results/evolved6502/coverage.csv`]

Each admitted role receives all 255 alternatives at every predicted free and erased coordinate and 64 joint assignments, against zero plus 16 uniform-byte partners. Eight successive fresh-partner passages compare every physical offspring byte and source label, without cyclic realignment or a whole-tape similarity gate. The final report records 91,428 intervention cases and 731,424 serial observations with no mismatch. Joint assignments are finite, not an enumeration of every product-family sequence. A separate descriptive serial scan takes two first and two final roles per arm, with 32 alternatives per coordinate, four mutant lineages and 64 controls; it uses marker thresholds and does not test the exact law. Its sites can include alternative machinery alleles.

[Generated exact and descriptive outputs: results/evolved6502/exact_cases.csv.gz, results/evolved6502/exact_serial.csv.gz, results/evolved6502/law_verdict.csv, results/evolved6502/descriptive_panel.csv, results/evolved6502/descriptive_mutants.csv.gz, results/evolved6502/RETENTION.md]

Registered Z80 displacement interventions

The displacement study contains 22 archived parents and 101 displacement settings, including 22 original-value controls and 79 changed settings; repeated tape strings and parent-world origins remain in the ledger. Every setting qualified before retention. Sixteen serial lineages follow each forward setting through generation eight; reciprocal passages restore the original displacement in actual descendants and repeat the transfer. This gives $101\times16\times8\times2 = 25{,}856$ serial rows. The canonical offspring, rather than the possibly nonperiodic archived parent tail, is the invariant family. [Generated settings: results/capacity/parents.csv, results/capacity/edit_eligibility.csv, results/capacity/serial.csv.gz]

Direct assays use actual generation-eight descendants. They test all 255 alternatives at every free and erased coordinate and 64 joint assignments against all 17 frozen partners. Identical descendants share an execution result, but their lineage identities remain separate. The 1,067,608 direct cases and all serial observations agree with complete byte and ancestry predictions. A case is a declared intervention, not one independent world or necessarily one encounter. Eligibility came from code and ordered copying, never a fit to measured site counts; no post-outcome exclusion was used. [Generated outcomes: results/capacity/capacity_cases.csv.gz, results/capacity/CAPACITY.md]

Constructed 8086 L1 and the empty registered evolved tests

The constructed cores are `bf dLo dHi 49 f3 a4` and `bf dLo dHi 49 f3 a5`: `MOV DI,d; DEC CX; REP MOVSB/MOVSW`. Reset requires zero DS, ES, SS, CS and SI, DF = TF = 0 and CX = 0; decrementing CX gives 0xFFFF. The baseline parent is d-periodic with core at physical entry zero. Require $6 \le d \le L$, d dividing L, even d for MOVSW, forward constant displacement, preserved core, no interfering writes, no original-partner reads and enough ordered copying to cover B. REP retains its decode and has budget-2 copying iterations, each counted against the budget and below count exhaustion. Exactly positions $6,\ldots,d-1$ are free; initial positions $d,\ldots,L-1$ are erased. The five L/d geometries give free counts 2 and 10 at L16 and 2, 10 and 26 at L32. [Frozen contract: results/census8086/LAW_CONTRACT.md; generated eligibility: results/census8086/law_eligibility.csv, results/census8086/law_geometry_traces.csv]

The preceding fixed-input census qualifies 340 geometry/operation/budget/partner rows. The S2 constructed control panel uses ten byte/word geometries at budget 128 and checks all 256 values at each free or mutable erased coordinate, 256 counterbalanced plus 256 seeded joint assignments, the 17 census partners and eight serial fresh-partner generations for single/joint cases. Core, physical phase and full offspring ancestry are scored, with a mismatch stopping diagnosis rather than being averaged away. All ten controls completed without mismatch in 1,185,792 encounters. L32 controls remain fixed-input evidence after withdrawal of the L32 population arm. [Generated controls: results/census8086/table_law.csv; results/s8086/tables/constructed_controls.csv, results/s8086/tables/control_hash_verification.csv]

Registered evolved candidates comprise first detection, first confinement, final top three and 64 uniformly sampled final cells per world; duplicate cell multiplicity is preserved. Each law family separately permits at most 16 eligible genotype-world cases per world, prioritizing first-detection, confinement and final roles before keyed random cells. L1 uses only the verified six-byte REP family; amended L2 uses Supplementary Note 2's broader certificate. Unresolved proof obligations suspend support; a reproducible mismatch that remains independently eligible would fail the application. [Sampling and rules: `REVISION_PREREG.md`, S2; generated ledger: `results/s8086/tables/law_candidates.csv`, `results/s8086/tables/law_worlds.csv`]

The registered ledger has 1,366 source records, 1,337 distinct genotype-world candidates and 779 culture-positive candidates. L1 admits none. L2 excludes 55 for original-partner reads and leaves 724 unresolved, including 708 at unmodelled `POP ES`. No family cap removes a case. Both registered evolved assay denominators are zero. L1 is untested; L2 is untested with support suspended. The frozen L2 reporter's `incomplete/invalid` certificate status does not invalidate the complete route cohort and is not an observed law mismatch. The post hoc extension in Supplementary Note 9 leaves those verdicts unchanged. [Generated denominators and empty outcomes: `results/s8086/tables/law_denominators.csv`, `results/s8086/tables/law_eligible_cases.csv`, `results/s8086/tables/law_outcomes.json`; `results/s8086/S2.md`]

Supplementary Note 5. Cohorts and measurements retain distinct denominators

Initialization, detection and sampling

The four measurements are culture, execution confinement, partner inflow and transfer retention; the two properties are partner-information closure and parental-variant transmission. These definitions do not supply additional experimental denominators.

Emergence worlds begin with pseudorandom byte tapes from the declared seeds; packed padding is excluded from the tape. Competition worlds instead begin with the prescribed spatial mixtures of partial-copy and full-tape designs. Nominal lattice worlds use 20,000 cells, four neighbours with reflecting boundaries and 8,192 drawn pair slots per update. Pair collisions mean that the 8,192 drawn slots do not guarantee the same number of active encounters. At noise exponent 4, the usual mutation setting attempts 512 uniformly located byte replacements per update; unchanged replacements and the disclosed packed-word lost-update race mean that attempted, applied and changed counts differ. A seed identifies initialization and nominal streams, not bitwise replay of a nondeterministic parallel trajectory. [Execution: `algocell_exp/soup.py`, `algocell_exp/ZSOUP.md`; audit: `results/s6502/CORRECTION.md`; registration: `REVISION_PREREG.md`, S2]

Supplementary Table 1. Cohorts, schedules and selection. Historical full run inventories are generated in `results/review_r2/RUN_TABLE.md`; the table below identifies the cohorts used in the main figures and later tests. Historical rows retain their original scorer definitions.

Cohort	Worlds, seeds, length and horizon	Observation schedule and selection	Generated source / frozen definition
Historical Stage E, nominal unablated subset	Ten worlds per displayed length, seeds 101–110; budget 128; horizon 300,000	Explicit early samples 1, 2, 3, 5, 8, 13, 21, 34, then every 50 to 5,000 and every 500 thereafter; first qualifying top-three exemplar at share ≥ 0.005 in chronological record order	`results/stageE/assays.csv`; `manuscript/figures/out/v8/data/emergence_worlds.csv`; `assay_batch.py`, `make_conds.py`
Historical Stage G	Twenty worlds at each L = 16, 20, 50, 64, seeds 2001–2020; budget 128; L16/L50 horizon 300,000, L20/L64 horizon 1,000,000	Same early schedule; every 500 after 5,000 in short runs and 1,000 in long runs; original top-three culture selection; route interpretation combines historical syntactic control-flow and partner assays	`results/stageG/stageG/stage_g_runs.csv`; `results/stageG/FINDINGS.md`; `make_conds.py`, `stage_g.py`
Historical H/I/K/L/M	H: ten L16 mixed-pair worlds, seeds 3001–3010; I: ten L16 lethal worlds, 4001–4010; K: twenty L32 worlds, 5001–5020; L: ten each L16/L20, 6001–6010; M: twenty each L16/L32, 7001–7020	Early samples as above; H/I and M-L16 horizon 300,000, K and M-L32 1,000,000, L 10,000,000; later intervals 500, 1,000 or 5,000 respectively; Stage L missing-snapshot issue described in Supplementary Note 7	`results/review_r2/RUN_TABLE.md`; `make_conds.py`; per-stage generated assay tables
6502 S1 A–D	Twenty per arm, seeds 65001–65020 reused; all horizons 10,000,000. A: L16/128/faithful BRK; B: L32/128/faithful; C: L16/512/faithful; D: L16/128/halt-before-BRK	Top three every 50 to 2,000, every 250 to 20,000, every 2,500 to 1,000,000, every 25,000 thereafter; first culture-positive top-three tape at ≥ 0.005, largest share at first qualifying sample. Matching lengths share initial conditions	`results/s6502/v2/arm_comparison.csv`; `results/s6502/CORRECTION.md`; `results/evolved6502/coverage.csv`; `REVISION_PREREG.md`, S1
Coherent-fetch 8086 S2 A	Twenty L16 worlds, seeds 8086001–8086020, budget 128, horizon 3,000,000; B/L32 withdrawn before data	Initial sample plus S1 intervals through 3,000,000 updates: 585 samples/world; checkpoints at 0; 20,000; 100,000; 1,000,000; 3,000,000 updates. First qualifying top-three tape at ≥ 0.005, largest share then lexicographic tie-break; every scheduled sample retained for descriptive persistence	`results/s8086/tables/worlds.csv`, `results/s8086/tables/kaplan_meier.csv`; `results/s8086/S2.md`; final S2 amendments
L32 mutation-on displacement competitions	Per environment and core cohort: ten equal mixtures, five partial-copy R introduced at 1%, five full-tape T introduced at 1%; horizon 300,000	Frozen byte classifier at each saved sample; all 20 worlds retained, including complete loss of R+T; world medians and endpoint distributions	`results/offset/offset_switch.csv`; `results/tier2/scored/E/offset_switch.csv`; `manuscript/figures/out/v8/data/competition_endpoints.csv`
Historical/core-C1 lethality dial	Ten worlds at each p = 0, 0.01, 0.03, 0.1, 0.3, 1, seeds 9001–9010 reused; L16/budget128, horizon 300,000	Top three every 50 to 2,000, every 250 to 20,000 and every 2,500 thereafter; culture/fetch measures under each world's own p	`results/dial/dial_worlds.csv`; `results/tier2/T2_DZ.csv`; `REVISION_PREREG.md`, DZ

The detection share threshold of 0.005 is 0.5%, or 100/20,000 cells.

S1 specified 64 culture partners, whereas its cited frozen final-cell estimator used 32; top-tape cultures used 64. Its trace prose specified soup partners, whereas the frozen code used 16 uniform-byte partners. Corrected frozen32 final-cell classifications are primary. Complete 64-partner final-cell reclassification and first-tape soup-snapshot tracing are separately reported sensitivities. The latter uses the earliest saved snapshot at or after the unchanged first-detection time and is not an exact contemporary-soup reconstruction. Neither sensitivity changed the first-tape classifications or final-cell counts. Every sampling role remains in the 6502 retention study; sensitivity re-evaluations do not create extra cells. [Generated comparisons: `results/s6502/CORRECTION.md`; `results/evolved6502/coverage.csv`]

Culture assays measure normalized generation-two gain relative to the original parent, including all first-generation outputs, even those that failed to become close copies. The score is not conditional on a successful first copy. S2's 64-partner generation-two rule and lexicographic tie-break are stated for S2; they are not retroactive replacements for historical selectors. Cross-machine curves compare recorded detections with different schedules and horizons, not exact births or a registered directional architecture effect. S2 has 14 detections, of which two are open and twelve confined; six worlds are right-censored at 3,000,000 updates. Its confinement endpoint has eight censored worlds and median 975,000. [Generated coordinates: `manuscript/figures/out/v8/data/arrival_worlds.csv`, `manuscript/figures/out/v8/data/arrival_curves.csv`, `manuscript/figures/out/v8/data/arrival_censors.csv`; `results/s8086/tables/kaplan_meier.csv`]

Survival and carriage are separate assays

At every historical serial or X1 branching transfer, an offspring qualifies only if at least 75% of its positions match its immediate parent at the best cyclic shift. A serial lineage survives through generation g only if all preceding transfers qualify. In X1, every surviving member meets two fresh partners and only qualifying offspring enter the next generation; parents are not carried forward. If more than 32 offspring qualify, 32 are sampled uniformly. A population survives if at least one member remains. X1 tests 32 seeded alternatives at every position, four starting populations per alternative and 64 unmutated populations through eight generations. Arm U uses uniform-byte partners; arm K uses the declared open-phase Stage G snapshot pools for pushers. [Registered design: `REVISION_PREREG.md`, X1; execution and outputs: `branching_transfer.py`, `results/branching/branching_summary.csv`, `results/branching/SUPP_TABLE.md`]

A population meets the carriage criterion when at least half of its members contain the introduced mutant's centred cyclic five-byte window. Identifiable alternatives exclude windows already present in the unmutated tape. Site scoring requires at least half of identifiable tested alternatives to meet the registered background-corrected criterion among surviving populations. Consequently the display's middle category is **alive with fewer than half carrying**, not demonstrated complete marker erasure. All mutant starts, rather than only survivors, form the plotted outcome denominator. Historical and corrected-core X1 endpoints agree for every plotted outcome and side-column count. The display therefore uses one bar per design while keeping the cohorts separate. For the displayed pusher, return closer, block-copy regenerator and L20 seed-6003 transmitter, control survivors are respectively 48, 64, 64, 64 out of 64, and scored sites are 0/16, 0/16, 0/32, 9/20. The nonzero regenerator carriage fraction is compatible with zero thresholded sites. [Generated plotting inputs: `manuscript/figures/out/v8/data/branching.csv`, `manuscript/figures/out/v8/data/branching_endpoints.csv`; corrected-core results: `results/tier2/scored/Branching/branching_summary.csv`]

Historical serial assays use 16 lineages per alternative and 256 unmutated controls. They exhaust all 255 alternatives for $L \leq 32$ and use 32 seeded alternatives for longer tapes. Reported observations are generations 1, 2, 4, 8; plotted fractions use all mutant starts. The execution and site strips mark fetched instruction-stream bytes, including operands. The transmitter in Figs 2a,b is L20 seed 6003; Fig. 2c uses L20 seed 6004. These specimens have different site counts. [Generated sites and summaries: `results/serial_retention/sr_sites.csv`, `results/serial_retention/sr_summary.csv`; `manuscript/figures/out/v8/data/serial_sites.csv`, `manuscript/figures/out/v8/data/serial_summary.csv`]

Fig. 2d follows the causal ancestry of one edited physical byte in an exploratory `ldir32` lineage. At zero-based execution step 6, it reads physical address 4 carrying original ancestor zero and writes `0x04` over marker `0xA7` at address 8, removing the last descendant of that edit. It is a complete erasure in this traced lineage, not the X1 carriage category. Every write is ordered; physical addresses 0–31 are A and 32–63 B. At cumulative execution step 128 the actual B output becomes A after reset and the absent ancestry remains absent through the next encounter. [Ordered trace: `manuscript/figures/out/v8/data/erasure_ordered.csv`, `manuscript/figures/out/v8/data/erasure_manifest.json`; full states: `manuscript/figures/out/v6/data/erasure_states.csv`]

Competition classes and figure measurements

The competition classes differ from the random-population site-score classes. At L32 the byte motif at positions 1–3 must be `5e ed b0`; byte zero is in R = {04,44,84,08,48,88,C8,50,90} or T = {60,A0}, with all entries hexadecimal. R includes partial-copy d = 8 and d = 16 designs with four and twelve predicted payload coordinates. T is the full-tape class, d = L. Other motif-bearing byte-zero values remain outside R and T. In random-population assays, by contrast, confined regenerators have at most two scored sites and confined transmitters at least five; intermediate cases remain separate. [Frozen classifier: `offset_switch.py`; generated census: `results/offset/OFFSET_CENSUS.md`; site-score classes: `results/ladder/LADDER.md`]

For each competition, T/(R+T) is a relative class share, while all core-bearing cells divided by 20,000 is motif prevalence. They have different denominators. Undefined ratios when R+T = 0 are omitted only from ratio summaries. The corrected-core benign T-into-R subgroup's reported median 0.08 uses four defined ratios among five worlds; the fifth remains at zero absolute abundance and zero tail diversity. Across that environment there are 19 defined ratios among 20 worlds. Distinct transmitter tails are nested diversity, not additional replicates. Historical and corrected-core trajectories retain their separate endpoints. Without mutation, E1's required stability interval was met in only four of ten corrected-core worlds at each tested length, short of seven; variation across worlds does not establish neutrality.

With mutation on, E2 required T/(R+T) to fall below its starting share and lie in 0.05–0.4 at 300,000 updates in $\geq$ 7/10 equal-start worlds. This criterion was met in only 2/10 L16 benign, 0/10 L16 lethal and 5/10 L32 benign worlds under the corrected core, compared with historical counts of 3, 0 and 6 of 10, respectively; E2 therefore failed in every registered condition in both cohorts. In the corrected-core L16 benign equal-start worlds, median T share was 0.70 but census-defined core abundance was only 0.05 of all cells. This loss limits interpretation of the relative share and differs from the L32 environmental contrast. [Generated endpoints: `manuscript/figures/out/v8/data/competition_endpoints.csv`, `manuscript/figures/out/v8/data/competition_medians.csv`; verdicts: `results/tier2/T2_COMPONENTS.csv`; E2 definition and historical outcome: `REVISION_PREREG.md`, E; corrected competition report: `results/tier2/scored/E/REPORT.md`]

The overlap schematic aggregates two bytes per displayed unit: L = 12, c = 4 and d = 4 or 12, with zero or eight free byte coordinates respectively. Repeated core labels mark ancestry copies rather than independently inherited positions. Fig. 3's exploratory population circles summarize 2,343 cells; motif/divisibility screening is not the full exact-law certificate. Its marked deficit at predicted four versus sampled zero comes from the whole-tape qualification gate; targeted transfers retain the predicted data. A different sampled exception, Stage K/L32/seed 5006, has six total scored sites, four unexecuted, with displacement eight; these counts describe the sampled site assay. The registered exact series are 101 Z80 settings and 42 6502 sampling roles, and the 296 constructed 6502 rows retain their sampled-carriage status. The registered figure contains no evolved 8086 exact-law point. [Generated displays: `manuscript/figures/out/v8/data/geometry_units.csv`, `manuscript/figures/out/v8/data/law_population.csv`, `manuscript/figures/out/v8/data/law_z80_c1.csv`, `manuscript/figures/out/v8/data/law_6502_c2.csv`, `manuscript/figures/out/v8/data/law_6502_controls.csv`; gate/score diagnoses: `results/crystal/cell_exceptions.csv`]

The two generation-eight intervention strips show parallel forward settings from one parent, not consecutive observations of a spontaneous mutation. Displacement is reduced modulo the encounter-ring length P = 2L = 64 at L32; byte `0x60` gives 96 mod 64 = 32. Byte `0x04` gives d = 4 and erases all 28 noncore coordinates at L32; `0x60` gives d = 32 and carries all 28. The exploratory 256-value byte-zero sweep has 157 nonworking values, 88 working values outside its recorded theorem conditions and 11 eligible working values. Its working threshold is two-generation normalized culture gain ≥ 0.3 on 16 uniform partners; this finite assay does not certify arbitrary partner or serial robustness. [Generated strips and sweep: `manuscript/figures/out/v8/data/lineage_strips.csv`, `manuscript/figures/out/v8/data/dial_coordinates.csv`; `results/crystal/alleles_ldir32.csv`]

The illustrative world in Fig. 1c is historical seed 2002, L16, with 20,000 cells. Modal match counts cells matching at least 12 of 16 bytes of the contemporaneous modal tape; zero fraction uses all bytes and distinct-tape fraction all cells. This is sequence similarity, not a spatial neighbourhood, a reproduction detector or a median world. Neither the assayed reference designs nor the return-containing conceptual design are asserted to descend from that world. Fig. 5d retains all ten first-family `5F A4 x y` tapes among fourteen S2 detections in their recorded physical phase; tile outlines group the four repeats, and positions are 0–15. The original descriptive labels use full instruction starts, whereas the later geometric explanation uses mapped ring positions. Colours are defined separately in each panel. [Generated display sources: `manuscript/figures/out/v8/data/world_trajectory.csv`, `manuscript/figures/out/v8/data/first_copier_tiles.csv`; conceptual source: `results/concept/traces.json`; measurement: `algocell_exp/metrics.py`]

The four serial-curve keys are `pusher16`, `ret16`, `ldir32` and `genome20_s6003`. The six byte-strip keys are `pusher50`, `ret16`, `ldir32`, `lethal16_s4009`, `genome20_s6004` and `closer32_8080_p1`. Hollow site symbols require transmissibility at generation one but not eight; filled symbols require generation-eight transmissibility. A never-fetched generation-eight site additionally has `executed=False`. The two lineage strips select `ldir32_v04` and `ldir32_v60`, lineage zero, generation eight. The illustrative marker trace has encounter indices one and two within one lineage, not two replicate lineages. Population-disc areas are proportional to cell counts, with three scale exemplars 1, 100 and 2,000; the 0–10 enlargement preserves coordinates. Byte-zero sweep entries run `00`–`FF` in hexadecimal. The emergence display begins at 40 updates, and the cross-machine arrival axis spans 40–12,000,000; these are display limits. Six S2 censoring observations coincide at one 3,000,000-update coordinate and are labelled as six censored worlds. These selections and limits are display definitions, not extra biological measurements. [Generated

coordinates: manuscript/figures/out/v8/data/serial_summary.csv, manuscript/figures/out/v8/data/serial_sites.csv, manuscript/figures/out/v8/data/lineage_strips.csv, manuscript/figures/out/v8/data/erasure_ordered.csv, manuscript/figures/out/v8/data/law_population.csv, manuscript/figures/out/v8/data/dial_coordinates.csv, manuscript/figures/out/v8/data/arrival_censors.csv; preparation/rendering: manuscript/figures/prepare_v7.py, manuscript/figures/prepare_v8.py, manuscript/figures/prepare_v62.py, manuscript/figures/build_v8.py]

Competition plots select three starting categories, `mix50`, `R_into_T` and `T_into_R`; reciprocal 1% invasions have initial T shares 0.01 or 0.99. The exploratory intruder comparison names two designs, R and T, and retains T_{tiled} as a separate control. The dial reports three outcome fields: open-first count, transmitter-majority count and confinement-time median. Transmitter majority requires more than 0.5 of heritable cells in the confined transmitter class. Arrival curves select three machine/cohort groups: Z80 G, 6502 A and 8086 S2. A claim component is one constituent registered criterion, and whole claims combine their component verdicts. These groupings do not supply extra independent experimental units. [manuscript/figures/out/v8/data/competition_endpoints.csv; results/offset/INTRUDER.md; manuscript/figures/out/v8/data/lethality_summary.csv; manuscript/figures/out/v8/data/arrival_worlds.csv; manuscript/figures/out/v8/data/arrival_censors.csv; results/tier2/T2_CLAIMS.csv; results/tier2/T2_COMPONENTS.csv]

Statistical procedures

Independent evolutionary units are worlds. Deterministic intervention mismatches are enumerated with zero tolerance; no confidence interval treats the many nested encounters as independent evolutionary trials. Kaplan–Meier summaries use scheduled event times and right censor complete nondetections at the declared horizon; the median is the first update where survival reaches one half and is unreached if it never does. Technically incomplete records are not biological nondetections. Competition summaries use world-level medians and retain undefined-ratio exclusions explicitly. Cumulative detection is 1 minus Kaplan–Meier survival.

I1 uses two-sided 95% Clopper–Pearson intervals and classifies a condition unaffected only when its upper bound is below 10^{-4}; there is no multiplicity adjustment, and the exploratory ring sweep is outside the decision (Supplementary Note 7). The lethality dial uses rank correlations with average ranks, 20,000 seeded permutations and the one-sided fraction of permuted statistics at least as large as observed; a printed zero means none of 20,000, not an exactly zero population probability. Its time-correlation input assigns nondetections the horizon, whereas its displayed medians use Kaplan–Meier censoring. The registered monotonicity/reversal condition is scored separately from correlation. [Code and generated scores: dial_soups.py, tier2_report.py; results/tier2/T2_DZ.csv, results/tier2/T2_COMPONENTS.csv]

The payload-toxin intervention averages each paired native-minus-control contrast over each host's encounters, then bootstraps hosts with replacement 4,000 times and reports percentile 2.5%/97.5% limits. This interval describes the sampled host panel. The L32 lethal native–detox intact-survival difference is +0.0009, 95% interval [0.0005,0.0013], below the registered explanatory threshold +0.02. Failure to reach that magnitude does not establish no protective effect. Exploratory population majority and class-detection order analyses remain descriptive; earlier detection is not ancestry. [Procedure: payload_toxin.py; generated contrast: results/toxin/CAUSAL.md; exploratory status: results/ladder/population_worlds.csv, results/ladder/LADDER.md]

Supplementary Note 6. Registration chronology and restart exclusions

Supplementary Table 2. Registration chronology and amendments. Dates and identities are archive facts, not reconstructed priority claims. The living registration retains historical superseded text; the dated amendments listed below identify the operative versions.

Date / stage	Frozen question or change	Outcome and scope	Authority and generated record
10 October, X1	Branching assay written before its code and runs; uniform U and open-soup K panels, separate survival and carriage predictions	X1-1 met, X1-2 failed in both partner arms, X1-3 met; core-C1 rerun retains these verdicts	`REVISION_PREREG.md`, X1 and outcome; `results/branching/BRANCHING.md`, `results/branching/SUPP_TABLE.md`; `results/tier2/T2_COMPONENTS.csv`
10 October, X3/N6	X3 question registered after discovery line records existed but before that analysis; N6 lineage replication prospective, with pre-run amendments	Discovery is not prospective confirmation. N6-1/2/3/5′ met; N6-4 failed; X3 confirmation met; genealogy definition changes remain recorded	`REVISION_PREREG.md`, N3 corrections, N6 amendments, X3; `results/lod_n6/N6_SCORE.md`; `results/lod/FOUNDING.md`, `results/lod/FOUNDERS2.md`
11 October, S1	Validated 6502 census and sizing pilot precede population registration; four arms frozen	Primary P1 not supported, P2/P4 supported, P3 overall untested because no eligible offset cells; its class component separately supported; no kill	`results/census6502/CENSUS.md`; `REVISION_PREREG.md`, S1; `results/s6502/v2/S1_v2.md`
11 October, S1 correction	General sequential-PC-wrap fix; 32/64 culture and uniform/soup trace conflicts disclosed; packed-word mutation race retained	All 80 fixed worlds rescored; endpoint/first classifications unchanged; no mutation-policy world rerun; primary frozen32 and sensitivities separate	`results/s6502/CORRECTION.md`, `results/s6502/v2/arm_comparison.csv`
11 October, I1	Core C0 equivalence and exploratory dry run precede confirmatory draw freeze at `bdcdcb6`; draw sets separated	All-unaffected criterion fails; omitted named Stage-E assay cases completed locally with sealed engine and separately recorded supplement	`REVISION_PREREG.md`, I1; `results/impact/exclusion.csv`, `results/impact/outcome.csv`, `results/impact/supplement_manifest.json`
11–12 October, I1-T2	New worlds/panels registered before execution, using frozen original designs/scorers and core C1; launch/scoring source `7821ff0`	19/22 claims retain all constituent verdicts; 4/36 components change; Stage-L complete/scorable despite dispatcher alerts	`results/tier2/T2_CLAIMS.csv`, `results/tier2/T2_COMPONENTS.csv`, `results/tier2/stageL_diagnosis.json`, `results/tier2/scoring_sources.json`
12 October, displacement C1	Registration `213a653`; eligibility/geometry frozen `4855ac3` before retention	101 settings in 22 archived parents; exact conditional prediction passes without post-outcome exclusions	`results/capacity/edit_eligibility.csv`, `results/capacity/CAPACITY.md`

Date / stage	Frozen question or change	Outcome and scope	Authority and generated record
12 October, 6502 retention C2	Registration `213a653`; 714-role extractor/ledger freeze `943aeee`; unchanged extraction expands to 942 roles at `657eb7e` before new interventions; descriptive inclusion clarification `f57ad24` precedes new assays	Complete 80-world panel; 42 eligible roles; exact conditional law passes; later C2 does not rescore earlier S1	`results/evolved6502/inputs.csv`, `results/evolved6502/coverage.csv`, `results/evolved6502/eligibility.csv`, `results/evolved6502/law_verdict.csv`, `results/evolved6502/RETENTION.md`
12 October, S2 scientific amendments before population data	Option (b) withdraws B/L32 population arm, retains A's 20 L16 worlds; source-root/core/physical-phase amendment repairs L2; observer and causal-classifier repairs frozen; prospective R2 withdrawal at `39de676`	No L32 population denominator or R2 confirmatory verdict. R0/R1/kill and L1/amended L2 remain prospective; constructed L32 controls remain fixed-input only	`REVISION_PREREG.md`, S2 B1–B6 and final amendments; `results/s8086/prelaunch/S2_REGISTRATION_OPTION_B.md`, `results/s8086/prelaunch/S2_FINAL_REFREEZE_2026-10-12.md`; final causal correction in Supplementary Note 8
12 October, S2 operational recoveries	Publication failure and restart from initialization, followed by scoring recovery	Scientific input, endpoint, sampling, eligibility and assay definitions preserved; failed attempts retained; completed worlds not rerun during scoring recovery	Dated operational amendments below; `results/s8086/provenance/run_provenance.json`, `results/s8086/provenance/verification.json`
12 October, S2 registered eligibility	Exact downloaded eligibility data committed at `8971d0` before evolved intervention stage	No certified case; registered L1/L2 assays empty, complete constructed controls; route cohort completes	`results/s8086/tables/law_denominators.csv`, `results/s8086/tables/constructed_controls.csv`; `results/s8086/S2.md`
12 October, S2 follow-ups	Post hoc dependency extension and purposive panel; predictions committed at `5bbc9d7` before that panel's assay	Exploratory admissions and exact matches do not revise registered verdicts; no new population world	`results/s8086/exploratory/law_panel_predictions.csv`, `results/s8086/exploratory/law_assay_denominators.csv`, `results/s8086/exploratory/publication_archive.csv`; `results/s8086/EXPLORATORY.md`

The genealogical discovery series changed definitions in response to identifiable failures of whole-tape and ancestry criteria. The correction ledger separates a confined founder from its writer, first founders from later conversions involving existing confined cells and whole-tape non-copiers from executed-code non-copiers. N6 tested the resulting frozen questions in twenty new worlds. Its founding-event and intermediary-chain components passed, but only eleven of twenty writers were non-copiers under the executed-byte criterion against twelve required; N6-4 remains unconfirmed. A detection ordering or abundant open ancestry does not by itself show a stepwise point-mutant transition. [Generated genealogy record: `results/lod/FOUNDING.md`, `results/lod/FOUNDERS2.md`, `results/lod/REACH.md`; `results/lod_n6/N6_SCORE.md`]

S2 restart and source continuity

World 8086001 completed computation but failed to publish its output; before reading biological output, the attempt was designated noncanonical and retained by opaque hashes. After the publication failure was repaired, the same seed restarted from initialization. Checkpoints lacked full continuation state, so no trajectory was spliced. Nondeterministic restart does not promise the same trajectory. After all worlds completed, a scoring interruption was repaired and only scoring onward resumed, without rerunning completed worlds. These changes preserved scientific source, machine, sampling, eligibility and endpoint definitions. [Dated records: `results/s8086/prelaunch/S2_OPERATIONAL_AMENDMENT_2026-10-12.md`, `results/s8086/prelaunch/S2_VOLUME_PARALLEL_AMENDMENT_2026-10-12.md`, `results/s8086/prelaunch/S2_SCORING_ONWARD_LOCK_AMENDMENT_2026-10-12.md`; `results/s8086/provenance/run_provenance.json`]

Source, world/checkpoint identities and constructed-control hashes were verified against the archived manifests; unchanged reporters generated the final verdicts. The manifests bind the prior execution freeze to each exact completed world's hash and retain the excluded attempt's identity. Failed attempts, restarts and operational amendments remain documented in the provenance archive. [Generated accounting/verification: `results/s8086/provenance/final_spend.json`, `results/s8086/provenance/verification.json`; `results/s8086/tables/control_hash_verification.csv`]

Operational incompleteness does not count as a biological negative. Arm reduction, R2 withdrawal and the source-root correction were scientific amendments before population data; subsequent restart and scoring recovery preserved those definitions.

Supplementary Note 7. I1 separates emulator impact from ecological replication

I1 compares historical compatibility core C0 with the corrected core and the vendor-specific Zilog-NMOS core. It executes 787 archived/random conditions with 100,000 confirmatory pairs each, or 78,700,000 pairs per core. The confirmatory stream excludes exploratory dry-run draws, with zero overlap in the generated exclusion check. The decision set has 690 conditions; the remaining ring-sweep conditions are reported descriptively. Final-pair-memory changes are measured under each condition's archived budget, mask and rule. Stochastic dial and random-register/stack settings use executor case-index streams, not the original ecological encounter streams. [Generated sources: `results/impact/conditions.csv`, `results/impact/metrics.csv`, `results/impact/exclusion.csv`, `results/impact/cores.csv`; frozen registration: `REVISION_PREREG.md`, I1]

For a changed-memory count x among n = 100,000 draws, the two-sided 95% Clopper–Pearson interval uses the corresponding binomial-tail bounds (equivalently beta quantiles, with exact zero/one edge conventions). A condition is unaffected only when its upper bound is $< 10^{-4}$, equivalent here to at most three changed pairs. This is a confidence-bound classification, not a statement that every memory was identical. There is no multiplicity correction. The complete interval audit checks every published row against independent binomial-PMF recurrences. The corrected core flags 47/690 conditions and the vendor-specific Zilog-NMOS core 114/690; their unaffected counts are 643 and 576. All 47 corrected-core flags and their claim links remain in the generated flagged ledger. [Generated intervals and decisions: `results/impact/rates.csv`, `results/`

impact/decisions.csv, results/impact/flagged_claims.csv, results/impact/interval_validation.csv]

The isolated-assay catalogue covers 417 tape/world/mask combinations and all fourteen output fields; 407/417 combinations are identical under each correction comparison. Complete score dictionaries, NaN equality and named-source aliases are retained without numeric tolerance. The original detached catalogue omitted named Stage-E unit-tiling cases; 119 additional source-table cases were completed locally using the same sealed engine and machines, with overlaps recorded rather than counted twice. All 126 cited source-table cases were checked under C0, with 630 published score comparisons and no mismatch. The original worker manifest and the omitted-case completion record are both retained. [Generated completion record: results/impact/assay_identity.csv, results/impact/assay_scores.csv, results/impact/assay_changes.csv, results/impact/assay_tapes.csv, results/impact/supplement_manifest.json, results/impact/unit_table_c0_validation.csv]

Affected encounters are attributed to the latest propagating repaired event: wasted prefixes with `LD A,R`, undocumented `BIT (HL)` flag behaviour, `SCF/CCF` vendor-latch behaviour or repeating-block instruction flags. Coherent F/Q intervention and exact source replay avoid assigning a coincidental equality to a mechanism. An unattributed "other" would stop the study; no such case occurred. These findings establish dependence on execution semantics and fail the all-unaffected criterion; they do not identify how an evolved soup's outcome would have changed. [Generated attribution: results/impact/attribution_classes.csv, results/impact/attribution_totals.csv, results/impact/affected_pairs.csv, results/impact/stop.csv, results/impact/outcome.csv]

I1-T2 reran the registered original ecological designs and seeds under the corrected core and rescored them with their frozen scorers. A claim stands only if every registered component equals its earlier verdict, including a prior failure. Nineteen of twenty-two claims stand; four of thirty-six components change. DZ2's reversal criterion and DZ3's high-lethality transmitter-majority criterion change from not met to met. Stage-L L2 changes at both lengths: L16 becomes inapplicable because no recovery occurs, and L20 becomes not met when a recovery appears without the required block/unexecuted property. The joint Stage-L kill still fires. E1 and E2 remain not met; the historical and corrected-core E2 counts are reported in Supplementary Note 5. C4b's failed literal motif-majority criterion remains separate from successful descendant confinement and reverse-pusher suppression. No-direction registrations and the Stage-E L10 descriptive addendum are not counted as positive predictions. Fresh nondeterministic worlds do not isolate a causal effect of core repair. [Complete generated verdicts: results/tier2/T2_CLAIMS.csv, results/tier2/T2_COMPONENTS.csv, results/tier2/T2.md]

Stage L is complete and scorable in all twenty worlds, while dispatcher completion remains zero of twenty because all end checks requested an update-7,500 snapshot absent from the sampling grid. The original archives likewise lack that snapshot. Both original and rerun schedules sample every 50 to 5,000 and every 5,000 thereafter; merging this grid with the explicit early samples jumps from 5,000 to 10,000. All journals reach 10,000,000 updates, final soups have the correct full size, and all twenty-three originally saved snapshots and every reference/final row needed by frozen scorers are present. No missing snapshot was synthesized, historical soup substituted or failed completion marker changed. [Generated diagnosis: results/tier2/stageL_diagnosis.json; retained scorer outputs: results/tier2/scored/; source identities: results/tier2/scoring_sources.json]

Instruction-vector agreement is a different validation level. The 8088 hardware vector record has 3,126,917 raw passes among 3,137,000 cases, with 10,083 expected hidden-latch divergences and none unexpected. Queue state and timing lie outside the coherent-fetch experiment. Z80/6502 reference vectors are emulator-generated; a same-core CPU comparison is not an independent hardware oracle. The five rows in Fig. 1d therefore retain separate denominators: architectural vectors, ring conditions below a change-rate bound, identical isolated assays, unchanged claims and unchanged components. [Generated display: `manuscript/figures/out/v8/data/validation_fractions.csv`; validation contracts: `algocell_exp/machines/PROVENANCE_8086.md`, `algocell_exp/machines/PROVENANCE_6502.md`; `results/emulator/RING_DIFFTEST.md`]

Supplementary Note 8. The causal census correction changes labels, not outputs

The validated 8086 census enumerates all 256 period-one words and all 65,536 period-two words at L16/L32 and budgets 128/512, against the zero partner plus sixteen frozen random partners. Constants and opposite phases remain in the enumeration, with no unevaluated words. The final descriptive correction audits 1,246 candidate settings across all seventeen contexts, giving $1{,}246 \times 17 = 21{,}182$ encounters. Raw memory, complete original machine records and archived trace bytes agree in every replay. Exactly 1,247 labels change relative to the preceding correction: 1,175 literal-writer labels and 72 memory-fed one-pass labels become "other". Output tapes, matches, coverage, access and raw counts remain unchanged. [Generated audit: `results/census8086/table_main.csv`, `results/census8086/descriptive_correction_labels.csv`, `results/census8086/descriptive_correction_counts.csv`, `results/census8086/descriptive_correction_provenance.json`; `results/census8086/DESCRIPTIVE_CORRECTION_2026-10-12.md`]

The operative classifier examines the episode through its last partner write. Every claimed store, including a numerically neutral read-modify-write, must pass isolated source substitutions `0x55` and `0xAA`. Instruction-start, instruction-fetch, data-read and write addresses must remain identical through the tested write, and its value must follow the claimed source expression. Later structural divergence is permitted. Literal claims require verified admitted operand/reset provenance; memory-fed claims require paired ordered reads and stores with the correct source-event identity. A confined copy-loop label additionally requires repeated full instruction-start addresses and no partner fetch in the classified episode. A repeated mapped ring position alone is insufficient. Mixed channels, unmodelled dependencies, failed substitutions, vector dependence and trace overflow remain "other", with reasons retained. Neutral writes still contribute coverage but cannot supply a new writing channel. Trace-only classifications and neutral-store shortcuts do not satisfy this contract. [Final contract: `results/census8086/MECHANISM_CONTRACT.md`, final prospective-correction section; intervention archive: `results/s8086/prelaunch/descriptive_intervention_checks.jsonl.gz`, `results/s8086/prelaunch/descriptive_random_validation.json`]

The corrected period-two nonzero exact-zero word counts are 20, 29, 26, 26 for L16/128, L16/512, L32/128 and L32/512. Their literal/memory-fed/confined-loop/other partitions are respectively 3/2/0/15, 3/1/0/25, 3/5/0/18 and 3/4/0/19. No nonzero period-one exact-zero word appears. These labels describe each audited encounter, not an establishment probability or a general capacity theorem. The witness `07 a5` remains verified memory-fed in all 68 settings/contexts, including all 64 random-partner contexts with partner instruction-stream fetch. At budget 128, `b9 51` bootstraps against zero and copies exactly into 13/16 random partners at L16 and 6/16 at L32;

being a literal load/push design does not override a failed causal classification in another context. `ff 30` and `ad 50` are not exact-zero bootstrappers under this reset. [Generated counts and witness: `results/census8086/table_hits.csv`; complete word inventories: `results/census8086/words_L16_k2_b128.csv`, `results/census8086/words_L32_k2_b128.csv`; `results/s8086/tables/census_07a5.csv`; `results/census8086/DESCRIPTIVE_CORRECTION_2026-10-12.md`]

Repeated changes to the classifier undermined the basis of R2's numerical forecasts. R2 occurrence and persistence were withdrawn as confirmatory predictions before any S2 population data, while the complete correction proceeded without a label-change stopping rule. Full-encounter fetch-based R1, culture-based R0, route kill and law proof obligations remained separate and unchanged. Full-encounter partner fetch and pre-first-write access are useful separate observables; neither is a substitute for this causal classification or for a complete heredity certificate.

Supplementary Note 9. S2 follow-ups are exploratory and post hoc

Missing pushers suggest a bounded establishment question

The scans search both physical phases of census-supported tiles in all 11,700 scheduled top-three sample records, containing 35,100 ranked tape entries and 2,000,000 checkpoint-cell observations: twenty worlds $\times$585 scheduled samples, three ranked tapes per record and twenty worlds $\times$five checkpoints $\times$20,000 cells. They count repeated observations, not unique lineages. No complete `b9 51`/`51 b9` tile or linear alternating run of at least four bytes is observed. The scan also records isolated pairs and other census witnesses. Absence at these observations cannot establish absence of unseen transient births or estimate a birth rate. [Generated scan: `results/s8086/exploratory/motif_scan.csv.gz`, `results/s8086/exploratory/motif_hits.csv.gz`, `results/s8086/exploratory/scan_contract.json`, `results/s8086/exploratory/snapshot_composition.csv`]

The encounter panel samples 512 distinct cells without replacement per world/checkpoint and uses matched seeded uniform partners. Tiles are planted either as focal executor A or receiving B, using budget 128 and physical entry A. Active-copy success accepts either exact tile phase in B; receiving survival requires the original planted B bytes at their physical phase. At 20,000 updates, `b9 51` copies into soup partners in 75.62% of challenges versus 53.20% into uniform partners, but survives in the receiving role in only 2.36% versus 25.66%. The opposite `51 b9` phase instead has active-copy success 32.71% versus 54.28%. Each pooled entry has $20\times512 = 10{,}240$ encounters nested within twenty worlds. The phase and role dependence motivates an arrival/establishment interpretation, not a measured growth rate or a historical causal explanation. [Generated fates: `results/s8086/exploratory/encounter_fates.csv`, `results/s8086/exploratory/encounter_fates_raw.csv.gz`]

The comparison Z80 panel uses all twenty Stage G L16 seeds 2001–2020 and snapshots at 500; 1,000; 2,000; 20,000; 100,000 updates. Early receiving-pusher survival is higher in this panel, but Stage G already contains expanded pushers by its first sampled time, whereas S2 lacks dense checkpoints before 20,000 updates. Machine semantics, mutation, observation times and histories differ; these data establish context dependence without isolating architecture or cause. Pre-existing and newly produced copies remain separate columns. [Generated cross-cohort identities: `results/s8086/exploratory/stageG_input_hashes.csv`, `results/s8086/exploratory/encounter_fates.csv`; interpretation: `results/s8086/EXPLORATORY.md`]

The four-byte first family transmits constrained machinery alleles

Ten of the fourteen registered first copiers are four repeats of `5f a4 x y`. At reset SP = 1, so `POP DI` reads A[1] = `a4` and A[2] = x and sets DI = 0xXXA4, congruent to four modulo 32. SI = 0 and DS = ES = 0; `MOVSB` copies forward at displacement four. Each traced exemplar makes 64 byte copies in 128 execution steps and writes every physical A/B coordinate. The final ancestry is [0,1,2,3] repeated. The conservative fixed footprint has R = K = C_R = {0,1,2,3}, E = {4,…,15} and F empty. This family has no free payload under that held four-byte footprint.

The x/y bytes supply a taken transfer under frozen reset flags, including NMOS `0x60–0x6f` conditional-jump aliases, ordinary conditional branches and `LOOP`. The `e9 3c` exemplar additionally fetches physical byte four, already overwritten by the copier, as part of its displacement. Full IP can advance or wrap while mapped execution stays in A; this does not change the original full-start-address mechanism labels. Ordered reads, writes, fetches and reconstructed pointers for first exemplars are retained, with a separate observer recording actual full SI/DI for the all-world panel. [Generated geometry: `results/s8086/exploratory/first_family.csv`, `results/s8086/exploratory/first_family_traces.csv.gz`, `results/s8086/exploratory/first_family_copies.csv`, `results/s8086/exploratory/family_world_traces.csv.gz`, `results/s8086/exploratory/family_world_copies.csv.gz`]

The descriptive founder-alternative assay varies each of the four founder coordinates through all 256 values, repeating the altered founder in all four homologous tiles. Each assignment faces sixteen fresh uniform partners and eight serial fresh-partner encounters at the physical phase. Across ten genotypes there are 10,240 assignments, including 10,200 non-baseline alternatives; 245 alternatives pass every direct and serial challenge. Every genotype has at least seven transmitted y alternatives, nine have nineteen x alternatives, and four have one byte-zero alternative. This establishes constrained heritable variation in executed machinery while leaving the fixed-footprint free count zero. These homologous founder substitutions differ from the exact law's single-physical-coordinate erasure tests. [Generated all-value outputs and failures: `results/s8086/exploratory/family_transmission.csv`, `results/s8086/exploratory/family_serial.csv.gz`]

A separate descriptive panel selects an exact family exemplar from every world with a scheduled full-family tape. This includes eighteen of twenty worlds. All eighteen copy exactly against sixteen fresh partners; seven remain unjudged by the registered culture protocol because their scheduled abundance never reaches its 100-cell gate. They are not culture-negative. The observer's 306 lanes (18×17 contexts) match uninstrumented memory and machine records. This broader trace panel is outside both the 779-candidate extraction denominator and the 32-genotype law panel, and it does not revise registered detection times. [Generated coverage/verification: `results/s8086/exploratory/family_world_coverage.csv`, `results/s8086/exploratory/family_world_exemplars.csv`, `results/s8086/exploratory/verification.csv`]

Extended dependency modelling supports a separate law panel

The post hoc verifier adds conservative dependencies for previously unmodelled segment/register POPs, stack and segment addresses, frozen jump aliases, ALU taint, input-return values, LES/LDS, far JMP, byte immediates and exchanges. Arithmetic memory writes are not admitted as exact copies. Every culture-positive candidate is re-extracted against all seventeen original contexts. Of 779 candidates, 600 become eligible, 72 excluded and 107 unresolved. Among the 708 cases originally unresolved at `POP ES`, 587 qualify, sixteen become excluded and 105 remain

unresolved. L1 remains empty. The newly admitted geometries are 590 full-tape d = 16 copiers and ten d = 4 copiers, all with source start and physical phase zero. Full-tape cases do not thereby become smaller-seed overlap founders. [Generated certificates and reasons: `results/s8086/exploratory/eligibility.csv`, `results/s8086/exploratory/eligibility_counts.csv`, `results/s8086/exploratory/certificates.json`; verifier identity: `results/s8086/exploratory/compute_source_hashes.csv`]

The purposive intervention panel selects the smallest held footprint in each of twenty worlds, all ten first-family genotypes and two deterministic additional cases: 22 full-tape and ten four-byte genotypes in total. Its complete K/R/C_R/F/E/source-map predictions were committed at `5bbc9d7` before this assay, but extraction and selection remain post hoc. The frozen assay is reused with the extended verifier and 64 independent seeded joint assignments replacing the registered 512-joint panel. It tests all-value single/free and mutable-erasure substitutions against seventeen census partners and eight fresh-partner serial transfers for each single/joint assignment. No favourable rotation, post-mismatch exclusion or new world run is introduced.

All 32 cases complete with no mismatch in 2,224,640 encounters: 66,048 single assignments over 258 genotype-coordinate pairs, 2,048 joint assignments and 30,720 erasure assignments. The encounter total is $(66{,}048+2{,}048)\times(17+8)+30{,}720\times17 = 2{,}224{,}640$. For d = 4, F is empty, so the 64 joint assignments repeat baseline and add no payload evidence; the erasure tests remain substantive. For d = 16, E is empty and erasure controls are inapplicable. CPU audits reproduce output/input hashes, serial partner streams, source maps and all denominators. These finite post hoc agreements do not revise registered L1/L2, certify unresolved candidates or establish unlimited robustness. [Generated predictions and counts: `results/s8086/exploratory/law_panel_predictions.csv`, `results/s8086/exploratory/law_assay_summary.csv`, `results/s8086/exploratory/law_assay_denominators.csv`, `results/s8086/exploratory/law_encounters/`, `results/s8086/exploratory/verification.csv`]

The publication archive preserves 239 provenance files byte-for-byte or by deterministic gzip, with original/stored hashes and lengths. No new population worlds were run, and the registered verifier and verdicts were unchanged. Committed tables document the follow-ups. [Archive and report: `results/s8086/exploratory/publication_archive.csv`, `results/s8086/exploratory/input_hashes.csv`, `results/s8086/exploratory/compute_source_hashes.csv`, `results/s8086/exploratory/gpu_bursts.csv`; `results/s8086/EXPLORATORY.md`]

Supplementary Tables and generated-source index

Supplementary Tables 1 and 2 describe the cohorts and registration chronology. The following tables identify the complete generated records, including row-level exclusions and failures. CSV files and gzip-compressed files contain machine-readable tables; the corresponding result reports explain their fields. The availability statements and accession placeholders are in `manuscript/MAIN_v8.md`.

Table	Content and units	Generated source files
3	Complete registered verdict ledgers, with failures, withdrawals, untested cases and original/core-C1 components separate	`results/tier2/T2_CLAIMS.csv`, `results/tier2/T2_COMPONENTS.csv`; `results/branching/SUPP_TABLE.md`; `results/lod_n6/N6_SCORE.md`; `results/s6502/v2/arm_comparison.csv`; `results/evolved6502/law_verdict.csv`; `results/capacity/CAPACITY.md`; `results/s8086/tables/law_denominators.csv`, `results/s8086/tables/law_outcomes.json`; `results/s8086/S2.md`
4	Historical serial and registered branching outcomes, identifiable alternatives, sites, controls and partner environments	`results/serial_retention/sr_summary.csv`, `results/serial_retention/sr_sites.csv`, `results/serial_retention/sr_mutants.csv.gz`; `results/branching/branching_summary.csv`; `results/tier2/scored/Branching/branching_summary.csv`; `results/pe/pe_matrix.csv`
5	Z80 displacement parents, setting eligibility, ordered geometry, descendants, reciprocal passages and direct tests	`results/capacity/parents.csv`, `results/capacity/candidate_eligibility.csv`, `results/capacity/edit_eligibility.csv`, `results/capacity/geometry_traces.csv.gz`, `results/capacity/serial.csv.gz`, `results/capacity/capacity_cases.csv.gz`, `results/capacity/partners.csv`, `results/capacity/inputs.csv`
6	Complete evolved 6502 sampling roles, archive coverage, eligibility/exclusions, exact and descriptive tests	`results/evolved6502/panel.csv`, `results/evolved6502/coverage.csv`, `results/evolved6502/inputs.csv`, `results/evolved6502/eligibility.csv`, `results/evolved6502/geometry.csv`, `results/evolved6502/exact_cases.csv.gz`, `results/evolved6502/exact_serial.csv.gz`, `results/evolved6502/descriptive_panel.csv`, `results/evolved6502/descriptive_mutants.csv.gz`
7	Literal-channel layouts, symbolic/direct searches, finite counterexamples, witnesses and bootstrap replays	`results/literal_channel/pair_layouts.csv`, `results/literal_channel/minimal_symbolic_search.csv`, `results/literal_channel/direct_search.csv`, `results/literal_channel/finite_search.csv`, `results/literal_channel/finite_bound_counterexamples.json`, `results/literal_channel/minimal_witnesses.json`, `results/literal_channel/finite_witness.json`, `results/literal_channel/6502_finite_harness.json`, `results/literal_channel/verification.csv`
8	Validated 6502/8086 finite census, constructed law conditions and controls	`results/census6502/table_main.csv`, `results/census6502/table_hits.csv`, `results/census6502/table_closers_law.csv`, `results/census6502/table_closers_exact.csv`; `results/census8086/table_main.csv`, `results/census8086/table_hits.csv`, `results/census8086/law_eligibility.csv`, `results/census8086/table_law.csv`; `results/s8086/tables/constructed_controls.csv`, `results/s8086/tables/control_hash_verification.csv`

Table	Content and units	Generated source files
9	Complete I1 rate bounds, flagged claims, isolated assay changes, attribution and omitted-case completion	results/impact/rates.csv, results/impact/decisions.csv, results/impact/flagged_claims.csv, results/impact/assay_identity.csv, results/impact/assay_changes.csv, results/impact/attribution.csv, results/impact/attribution_totals.csv, results/impact/supplement_manifest.json, results/impact/interval_validation.csv
10	Environmental competitions, ratio/core-loss denominators, dial, rejected explanatory tests and exploratory class order	results/offset/offset_switch.csv, results/offset/offset_census.csv, results/offset/offset_flow.csv.gz; results/tier2/scored/E/offset_switch.csv, results/tier2/T2_DZ.csv; results/toxin/CAUSAL.md; results/n2/BACKUP.md; results/ladder/population_worlds.csv
11	Final causal-classifier correction, every changed/other case and current per-setting mechanism totals	results/census8086/descriptive_correction_labels.csv, results/census8086/descriptive_correction_counts.csv, results/census8086/descriptive_other_cases.csv, results/census8086/table_mechanisms.csv; results/s8086/tables/judged_encounters.csv.gz, results/s8086/tables/census_07a5.csv
12	Registered S2 worlds, censoring, descriptive R2, candidate multiplicity, exclusions and empty law denominators	results/s8086/tables/worlds.csv, results/s8086/tables/kaplan_meier.csv, results/s8086/tables/persistence.csv, results/s8086/tables/law_candidates.csv, results/s8086/tables/law_worlds.csv, results/s8086/tables/law_denominators.csv, results/s8086/tables/law_eligible_cases.csv, results/s8086/tables/law_outcomes.json
13	Post hoc S2 motif scans, encounter roles, first-family and all-world traces, admissions, predictions and exact-law checks	results/s8086/exploratory/motif_scan.csv.gz, results/s8086/exploratory/encounter_fates.csv, results/s8086/exploratory/family_transmission.csv, results/s8086/exploratory/family_world_coverage.csv, results/s8086/exploratory/eligibility_counts.csv, results/s8086/exploratory/law_panel_predictions.csv, results/s8086/exploratory/law_assay_denominators.csv, results/s8086/exploratory/law_assay_summary.csv, results/s8086/exploratory/verification.csv
14	Genealogy discovery/correction and prospective N6 first-founder versus later-conversion records	results/lod/founding.csv, results/lod/founders2.csv, results/lod/open_first.csv; results/lod_n6/founders2.csv, results/lod_n6/N6_SCORE.md; REVISION_PREREG.md, dated N3/N6/X3 changes
15	Figure input coordinates, measurement definitions and provenance manifests	manuscript/figures/out/v8/data/; manuscript/figures/out/v8/qa_manifest.json; results/s8086/provenance/artifact_manifest.json, results/s8086/provenance/verification.json; results/s8086/exploratory/publication_archive.csv

Extended Data display map

The Extended Data displays and their source mappings are listed below. The display manifest is `manuscript/figures/out/v8/ED_V8.md`.

Extended Data item	Display	Content and sources
1. Validation and correction impact	`manuscript/figures/out/v8/ed8_1.pdf`	Instruction-vector, ring, isolated-assay and ecological units; expected divergences, the I1 rate rule and all-unaffected failure, I1-T2 original/corrected verdicts and the Stage-L completeness issue. Sources: Supplementary Tables 3, 9, 15 and Supplementary Note 7.
2. Survival, carriage and erasure	`manuscript/figures/out/v8/ed8_2.pdf`	Serial and branching maps and controls; identifiable alternatives and partner panels; lost, alive with < 50% carriage and alive with $\geq$ 50% carriage categories; ordered ancestry distinguishing actual erasure. Sources: Supplementary Table 4, Supplementary Note 5 and the v8 erasure trace.
3. Inherited displacement and capacity	`manuscript/figures/out/v8/ed8_3.pdf`	All 101 settings/22 parents, forward and reciprocal descendants, exact source maps, joint and erased controls, independent-world origins and exploratory gate exceptions. Sources: Supplementary Table 5 and Supplementary Notes 1–4.
4. Sampling, reset and classification	`manuscript/figures/out/v8/ed8_4.pdf`	Modal versus random-cell denominators, reset and stack sensitivity, motif failures, classes and exploratory detection order. Ancestry is distinguished from temporal detection. Sources: `results/r4/`, `results/population/`, `results/marker_check/`, `results/ladder/` and Supplementary Table 1.
5. Environmental competitions	`manuscript/figures/out/v8/ed8_5.pdf`	Historical/core-C1 trajectories, absolute and relative abundance, core-loss and undefined-ratio cases, tails, failed E1 and E2 abundance predictions, rejected explanatory predictions, the small toxin effect and exploratory intruder interactions. Sources: Supplementary Table 10 and Supplementary Notes 5, 7.
6. Machine laws and finite censuses	`manuscript/figures/out/v8/ed8_6.pdf`	Complete 6502/8086 census scopes, final causal labels, the literal finite bound and counterexamples, constructed controls, the S1 correction and all C2 roles/exclusions. Pointer randomization remains unmeasured. Sources: Supplementary Tables 6–8, 11 and Supplementary Notes 3, 4, 8.
7. BFF literal and environmental controls	`manuscript/figures/out/v8/ed8_7.pdf`	Literal, pointer, bandwidth and halting controls; open fetch with zero inflow and retained failed registered components. Sources: `results/bff/`, `results/bff_dials/`, `results/biology/individuality/` and Supplementary Notes 2, 3.
8. Founder genealogies	`manuscript/figures/out/v8/ed8_8.pdf`	Discovery-definition corrections, first founders versus later conversions, functional writers and N6 confirmation with failed N6-4 retained. Sources: Supplementary Table 14 and Supplementary Note 6.

Extended Data item	Display	Content and sources
9. Registered S2 and separate follow-ups	`manuscript/figures/out/v8/ed8_9.pdf`	Routes and censoring, descriptive R2, the complete frozen candidate/exclusion/unknown ledger and empty law assays. Post hoc motif, fate, family, admission and intervention panels are identified separately. Operational attempts are provenance, not replication. Sources: Supplementary Tables 11–13 and Supplementary Notes 6, 8, 9.

Legends state denominators and registered or exploratory status; generated source tables and hashes document provenance. Fig. 2 distinguishes carriage below the threshold from complete erasure. Instruction-stream fetching includes operands. Competition R/T classes denote partial-copy/full-tape designs. The 6502 series contains 42 roles from 41 tapes and eleven worlds. The displayed 8086 law status refers to the registered untested assay.

a **Validation: distinct denominators**

Comparison / agreement criterion	Met / tested units	Not met, n
8088/hardware: identical vector states	3,126,917/3,137,000	10,083
Corrected/historical: below change-rate bound	643/690	47
Corrected/historical: identical assay metrics	407/417	10
Corrected/historical: unchanged claim verdicts	19/22	3
Corrected/historical: unchanged components	32/36	4

All 10,083 hardware divergences were expected

Stable: upper 95% CI of changed-final-memory pair fraction <0.01%
100,000 encounter draws/condition; units named in each row

b **Stability flags and assay differences**

Core	Flagged / conditions	Assays differing / tested
Corrected	47/690	10/417
C2	114/690	10/417

Comparator: historical compatibility core C0
Corrected core (C1): classic; core C2: Zilog NMOS
Overall stability verdict: criterion not met for either core

Stable: upper 95% CI of changed-final-memory pair fraction <0.01%
Flag = stability criterion not met; assays compare with core C0

c **I1-T2: all 22 claim verdicts**

Claim	vs archive	Claim	vs archive
DZ1	stands	C4b	stands
DZ2	changed	C4c	stands
DZ3	changed	F4c	stands
E1	stands	S1	stands
E2	stands	S2	stands
E3	stands	S3	stands
E-L32L	stands	S4	stands
L1	stands	S5	stands
L2	changed	X1-1	stands
L3	stands	X1-2	stands
L	stands	X1-3	stands

DZ2 timing; DZ3 transmitter trend; L2 recovery sites
Stands = same as archive; E1 stability remains failed; 19/22 stand

d **Four changed components of 36**

Component	Historical	Corrected core
DZ2	not met	met
DZ3	not met	met
L2 L16	not met	n/a
L2 L20	n/a	not met

n/a = not applicable: no recovered lineage
Component tests summarized in panel a; L = tape bytes
DZ2/DZ3: criteria now met; L2: applicability changes

Historical = published archive; DZ timing/share; L2 sites
Complete 36-component ledger retained in Source Data

e **Assay changes by measurement**

Metric	Differing corrected core	Differing C2
Partner-copy fraction, parent A	0	0
Partner-copy fraction, parent B	6	6
Initial byte-match fraction	0	0
Informative partners, generation 1 (n)	0	0
Informative partners, generation 2 (n)	0	0
Cyclic copy offset (bytes)	0	0
Parent-byte fraction preserved, A	0	0
Parent-byte fraction preserved, B	9	9
Offspring ≥75% parent-match fraction	0	0
Generation-2 normalized gain	0	0
Generation-2 conditional gain	0	0
Normalized culture gain	0	0
Generation-2 gain ≥0.3 (yes/no)	0	0
Faithful culture classification (yes/no)	0	0

Counts /417 assays vs core C0; rows overlap (not additive)
A/B = parent role; copy = ≥75% immediate-parent byte match

f **Stage-L scoring and snapshot availability**

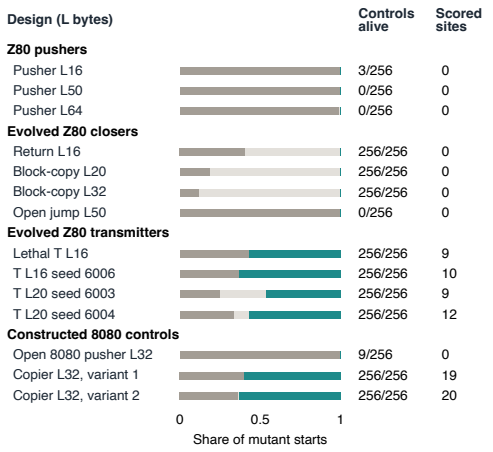
L bytes	Worlds, n	Scoreable from final data, n	Missing 7,500-update snapshot, worlds
16	10	10	10
20	10	10	10

All worlds completed 10,000,000 updates
Retained final bytes and journals support scoring
Applied-mutation/collision counts unavailable for these worlds

Stage L: closure-recovery worlds, 10 / tape length
The missing observation is a world snapshot at 7,500 updates

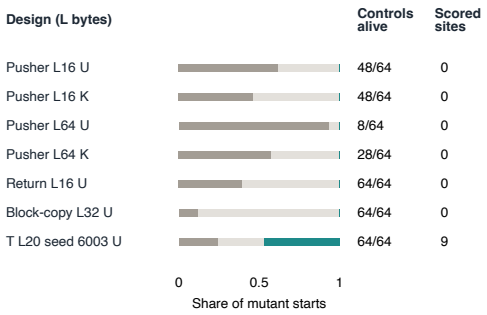
Extended Data 1. Validation and correction impact a, Validation comparisons identify their observational units and agreement criteria: instruction-vector states, ring conditions, isolated-assay metrics, claim verdicts and component outcomes. All 10,083 hardware divergences are expected. b, A ring condition is stable when the upper bound of its two-sided 95% Clopper–Pearson interval for the fraction of pairs with changed final memory is below 0.01%, using 100,000 encounter draws per condition. Flags indicate failure of that criterion. Historical core C0, corrected core C1 and core C2 identify emulator implementations, distinct from displacement experiment C1 and 6502 retention experiment C2. c, Nineteen of 22 I1-T2 claims retain every constituent verdict, including prior failures. d, Four of 36 components change; applicability changes are distinguished from newly met criteria. e, Counts of differing measurements use 417 isolated assays per core comparison; rows overlap. f, All ten Stage-L worlds at each tape length complete 10,000,000 updates and are scoreable from retained final bytes and journals, although their 7,500-update snapshots are missing. Applied-mutation and collision counts are unavailable for these worlds. Expected hardware divergences and stochastic rerun differences do not isolate a causal effect of core repair.

a **All historical serial designs: transfer 8**



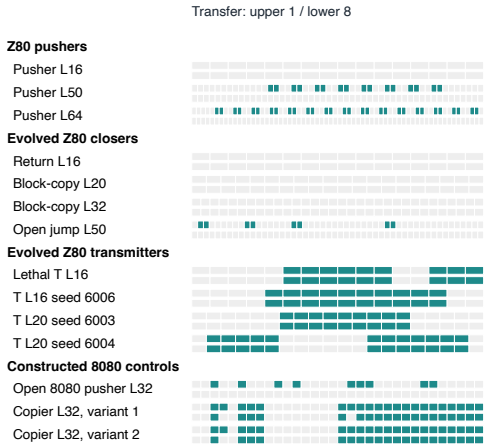
Dark grey lost; light grey alive noncarrier; teal alive carrier
Carrier = tape contains centred cyclic mutant 5-byte window
Site: ≥half identifiable tested values have excess carrier share ≥0.5
Excess = carriers/survivors (mutant) − carriers/survivors (WT)

b **Branching transfer-8 outcomes: U / K**



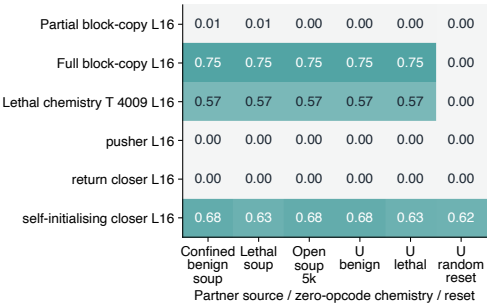
Grey: no surviving lineage; pale: alive, <50% members carry
Teal: alive, ≥50% members carry
Carriage = members containing mutant 5-byte window / all members
Site: ≥half identifiable tested values have excess carrier share ≥0.5
Excess as a; U = uniform-byte partners; K = evolved-soup partners
32 values×4 starts/site; WT = unmutated control (64)

c **Complete serial site maps: 1 / 8**



Upper: transfer 1; lower: transfer 8
Teal = scored site; pale = tested, unscored site
Site: ≥half identifiable tested values have excess carrier share ≥0.5
Same excess definition as a; tracks span bytes 0 to L−1

d **All partner-environment carriage tests**



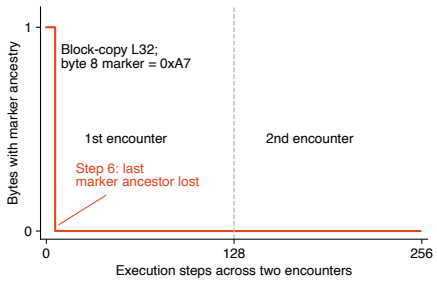
Transfer-8 carriers / all mutant starts; U = uniform-byte partners
All 6 parents L16: 16×255×16 = 65,280 starts/design; WT 256
Benign zero executes; lethal zero halts; open soup at 5,000 updates

e **Identifiable alternatives and controls**

Design	Identifiable / tested alternatives	WT starts
Z80 pushers		
Pusher L16	4080/4080	256
Pusher L50	1600/1600	256
Pusher L64	2048/2048	256
Evolved Z80 closers		
Return L16	4080/4080	256
Block-copy L20	5100/5100	256
Block-copy L32	8160/8160	256
Open jump L50	1600/1600	256
Evolved Z80 transmitters		
Lethal T L16	4080/4080	256
T L16 seed 6006	4080/4080	256
T L20 seed 6003	5100/5100	256
T L20 seed 6004	5100/5100	256
Constructed 8080 controls		
Open 8080 pusher L32	8160/8160	256
Copier L32, variant 1	8160/8160	256
Copier L32, variant 2	8160/8160	256

Identifiable: marker window absent from WT; T = evolved transmitter
Alternative types = L byte positions × tested values
16 starts/value; pusher50/64 & jump50: 32 values, others 255

f **Ordered ancestry: two full encounters**

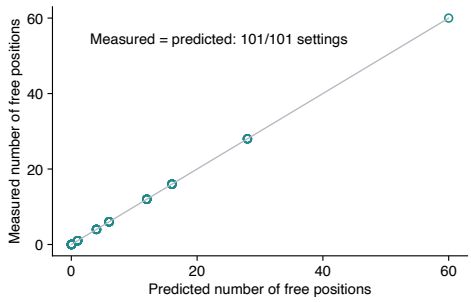


Counted unit: bytes with byte-8 ancestry; all-zero byte partners
Step 6 reads address 4 (ancestry byte 0), writes 0x04 over 8
At execution step 128: offspring becomes parent; processor reset

Extended Data 2. Survival, carriage and erasure a, Historical serial transfer-eight outcomes over all identifiable mutant starts: lost, alive without the mutant marker window, or alive with it. Each serial population contains one descendant. b, Registered branching transfer-eight outcomes distinguish loss, survival with fewer than half of members carrying the marker, and survival with at least half carrying it. The marker is the centred cyclic five-byte window of the introduced mutant. A scored site requires at least half of identifiable tested values to have an excess carrier share of at least 0.5: the fraction of surviving mutant

populations classified as carriers minus the corresponding fraction of surviving unmutated populations. Serial populations use the same criterion with one member. c, Complete serial site maps at transfers one and eight. d, Carrier lineages divided by all mutant starts for six L16 parents across partner environments; each design has 65,280 mutant starts and 256 unmutated controls. Benign zero opcodes execute; lethal zero opcodes halt. e, Identifiable alternative types and unmutated starts for the serial designs. f, One block-copy L32 lineage against all-zero byte partners, through two 128-execution-step encounters. At step six, a read from address four carrying byte-zero ancestry overwrites the final byte-eight marker descendant. Survival requires at least 75% immediate-parent byte match at the best cyclic shift; marker carriage and ancestry are separate measurements.

a **Registered: 101 settings (controls included)**



Free-position count: 0, 1, 4, 6, 12, 16, 28, 60
Settings, respectively: 32,14,10,14,10,14,6,1
Free byte positions: arbitrary alternatives jointly inherited with causal roots
Displacement experiment C1; fixed core; 22 parents /101 settings
Includes 22 original controls; markers coincide; diagonal = equality

b **Direct challenges: all passage directions**

Edit / challenge	Intervention cases	Passed
forward / erased	346,290	346,290
forward / joint	6,464	6,464
forward / single	181,050	181,050
reciprocal / erased	486,540	486,540
reciprocal / joint	6,464	6,464
reciprocal / single	40,800	40,800
Total	1,067,608	1,067,608

Z80 C1: 101 settings, 22 parents; actual transfer-8 descendants
Reciprocal = restore displacement; single = 1 site
Joint = payload jointly varied; erased = lost-coordinate control
Pass: exact offspring, ancestry and fixed core
Intervention counts; each tested against 17 partners

c **Z80 C1, L32: lineage 0, transfer 8**

Parent byte 0 = 0x04; copy offset 4 bytes




```
00–15: 04 5e ed b0 04 5e ed b0 04 5e ed b0 04 5e ed b0
16–31: 04 5e ed b0 04 5e ed b0 04 5e ed b0 04 5e ed b0
```

Parent byte 0 = 0x60; copy offset 32 bytes



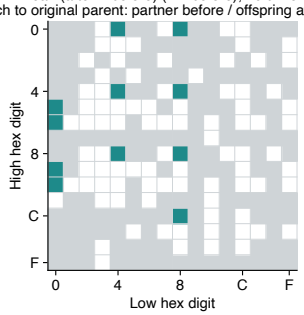

```
00–15: 60 5e ed b0 04 5e ed b0 04 5e ed b0 04 5e ed b0
16–31: 04 5e ed b0 04 5e ed b0 04 5e ed b0 04 5e ed b0
```

Fixed core Ancestry lost Ancestry retained

Byte positions 0–31; each example is one wrapped descendant
Ancestry key refers to original-parent byte sources

d **Exploratory: all 256 first-byte values**

Gain = mean (after − before)/(1 − before); 16 uniform partners
Match to original parent: partner before / offspring after; gen. 2



157: culture-negative (gain <0.3)
88: culture-positive (gain ≥0.3); prefix/offset/phase screen not met
11: culture-positive (gain ≥0.3); prefix/offset/phase screen met

e **Exploratory: sampled-score exceptions**

Measurement	Cells
Motif-bearing cells	2347
Cells with dividing offset	2343
Cells with extra fetched-site scores	53
Cells failing whole-tape match gate	1

53 and 1 are disjoint cell subsets of the 2,343 cells
Extra = fetched-site retention beyond unexecuted sites
Gate deficit: mutant alleles retained at 4 byte positions
Offspring <75% immediate-parent byte match at best cyclic shift

Archived Z80 cohorts G/I/K/L; 5e ed b0 motif (family as c)
2,347 cells from 42 worlds; 4 nondividing excluded

f **Frozen candidate eligibility**

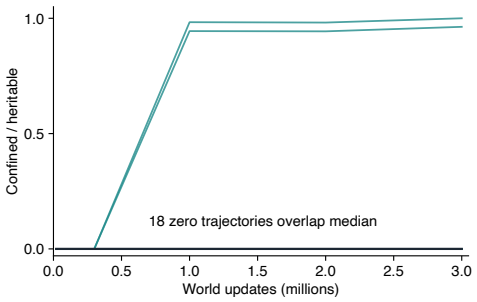
Eligibility / reason	Candidates
nondividing or out of range	4
Eligible	2343

Same motif-cell candidates as e: d divides L, 4≤d≤L
d = copy offset; L = tape length in bytes
Displacement experiment C1: 0/25,856 transfer-record mismatches

101 registered Z80 settings ×16 lineages ×8 transfers ×2 directions
Exact bytes, fixed core and ancestry: each 0/25,856 mismatches

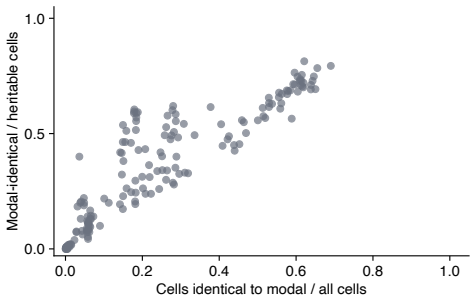
Extended Data 3. Inherited displacement and capacity a, All 101 registered displacement experiment C1 settings in 22 Z80 parents, including 22 original controls. Free byte positions admit arbitrary alternatives jointly inherited with their causal roots while the core is fixed; markers coincide at eight coordinates with setting counts shown. b, All direct single-site, joint-payload and erased-coordinate challenges, separately for forward and reciprocal passages, total 1,067,608 cases without mismatch; each case uses 17 partners. c, Actual transfer-eight descendant bytes and original-parent ancestry for parallel byte-zero interventions 0x04 and 0x60. d, The exploratory Z80 L32 first-byte sweep partitions all 256 values into 157 culture-negative, 88 culture-positive but theorem-ineligible and 11 culture-positive eligible values. Culture positivity requires dimensionless generation-two gain of at least 0.3 on 16 uniform partners; gain averages the increase in parent-byte match divided by available match headroom. e, The exploratory 5e ed b0 motif-bearing cell cohort from stages G, I, K and L; extra fetched-site scores and the immediate-parent whole-tape-match gate deficit are distinct outcomes. f, Frozen eligibility accounting for those 2,347 cells, and zero mismatches among 25,856 registered descendant-transfer records. Exact bytes, fixed core and ancestry each use that serial denominator. Eligibility and sampled scores do not certify a universal inheritance law.

a **Random-register reset: Z80 L16, 20 worlds**



At 3 million updates: 2/20 worlds majority confined
Thin = worlds; black median; cells confined / heritable cells
Confined: no partner instruction-stream fetch in 16/16 traces

b **Random cells versus the modal tape**



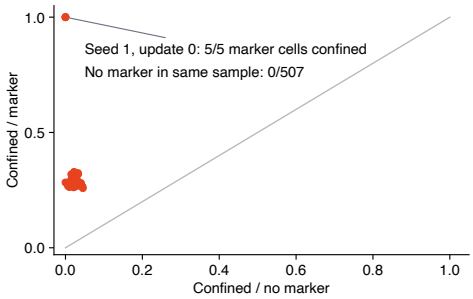
One dot = final world sample; 170 worlds in cohorts of c
Y: sampled heritable cells identical to modal / sampled heritable cells

c **Class denominators by cohort**

Cohort	Worlds	Cell rows	Confined rows	T rows
G, L16	20	3938	2624	259
G, L20	20	3757	2032	358
G, L50	20	5152	1165	19
G, L64	20	5444	897	320
I, L16	10	1217	1217	1008
K, L32	20	5519	741	149
L, L16	10	3210	2649	112
L, L20	10	2556	1783	253
M, L16	20	3966	2536	0
M, L32	20	5600	11	0

Pooled sampled-cell rows; confined and T are subsets
T = confined with ≥5 scored sites
G benign Z80; I lethal Z80; K: L32 ring aligned with 16-bit address wrap
L: closure-recovery extensions to 10 million updates
M 8080 subset; L in labels = tape bytes (not cohort L)

d **Sequence marker is not confinement**



Z80 L50 matched-pair invasion: 3 worlds, 24 samples
Sequence marker: 20 f0 (hex bytes); positive/negative cells
512 cells/sample; equal-fraction diagonal

e **Exploratory detection order: any detection**

Cohort	World count	R first	Same update	T first	R only	T only	None
G, L16	20	0	1	1	17	1	0
G, L20	20	4	9	0	3	3	1
G, L50	20	8	1	0	11	0	0
G, L64	20	18	0	0	2	0	0
I, L16	10	1	1	5	0	3	0
K, L32	20	1	8	2	5	0	4
L, L16	10	0	0	1	9	0	0
L, L20	10	4	4	2	0	0	0
M, L16	20	0	0	0	20	0	0
M, L32	20	0	0	0	8	0	12

R = confined ≤2 scored sites; T = confined ≥5 scored sites
R<T: R observed first; Same: same sampled update; T<R: T first
Any = positive estimated class share; n = worlds; not ancestry

f **Detection-order threshold sensitivity**

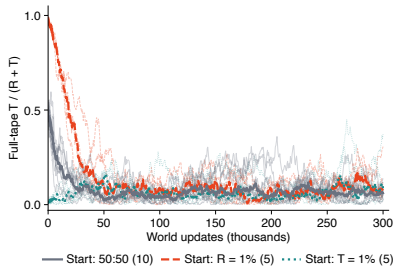
Threshold	R first	Same update	T first	Both seen	World count
G, L16 >50%	0	0	0	0	20
G, L16 >10%	0	0	1	1	20
G, L20 >50%	0	0	2	2	20
G, L20 >10%	6	5	0	11	20
G, L50 >50%	0	0	0	0	20
G, L50 >10%	1	0	0	1	20
G, L64 >50%	2	0	2	4	20
G, L64 >10%	8	0	0	8	20
I, L16 >50%	1	0	0	1	10
I, L16 >10%	1	1	1	3	10
K, L32 >50%	0	0	1	1	20
K, L32 >10%	1	5	3	9	20
L, L16 >50%	0	0	1	1	10
L, L16 >10%	0	0	1	1	10
L, L20 >50%	0	0	2	2	10
L, L20 >10%	5	3	2	10	10
M, L16 >50%	0	0	0	0	20
M, L16 >10%	0	0	0	0	20
M, L32 >50%	0	0	0	0	20
M, L32 >10%	0	0	0	0	20

R = confined ≤2 scored sites; T = confined ≥5 scored sites
Threshold: R or T cells / heritable cells
Both detected = sum of the three order columns; n = worlds
Site: ≥half tested byte values heritable, allele in ≥50% qualifying copies

Extended Data 4. Sampling, reset and classification a, A separate random-register-reset cohort of 20 Z80 L16 worlds, with six observation updates. Confinement fractions use sampled heritable cells as denominator; two worlds are majority confined at 3,000,000 updates. b, Final-world modal-tape abundance versus modal identity among sampled heritable cells in the 170 worlds of panel c. c, Sampled-cell and world denominators by stage and tape length; confined and transmitter cells are subsets of the pooled cell rows. d, The Z80 L50 matched-pair invasion cohort comprises three worlds and 24 samples of 512 cells. Axes compare confinement among cells with and without the 20 f0 sequence marker; seed one at update zero has

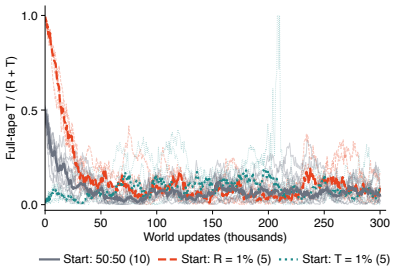
five of five marker cells confined and zero of 507 nonmarker cells confined. e, Exploratory order of first regenerator and transmitter detections at any positive estimated share. f, Complete sensitivity to majority and greater-than-10% class shares among heritable cells. Both-detected worlds equal the sum of the three mutually exclusive order columns. Regenerator and transmitter here mean confined site-score classes with at most two and at least five sites, respectively; they differ from competition byte classes. Observation order does not establish ancestry.

a **Historical core C0 / benign: L32, mutation on**



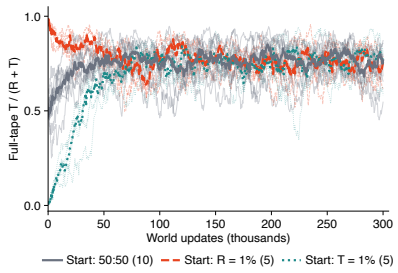
Parentheses count worlds. Thin: worlds; thick: medians
R partial-copy / T full-tape
Benign chemistry: zero executes; lethal chemistry: zero halts

b **Corrected core / benign: L32, mutation on**



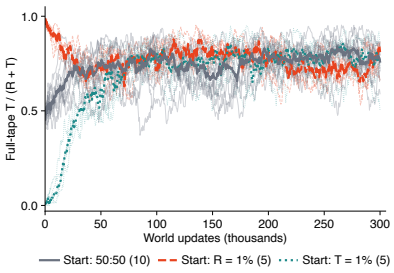
Parentheses count worlds. Thin: worlds; thick: medians
R partial-copy / T full-tape
Benign chemistry: zero executes; lethal chemistry: zero halts

c **Historical core C0 / lethal: L32, mutation on**



Parentheses count worlds. Thin: worlds; thick: medians
R partial-copy / T full-tape
Benign chemistry: zero executes; lethal chemistry: zero halts

d **Corrected core / lethal: L32, mutation on**



Parentheses count worlds. Thin: worlds; thick: medians
R partial-copy / T full-tape
Benign chemistry: zero executes; lethal chemistry: zero halts

e **Complete endpoints: 110 worlds/core; C1 Corrected, C0 Historical**

						T share		Core share		Tail types	
L bytes	Zero rule	Mutation	Start	Worlds Corr/H	C1	C0	C1	C0	C1	C0
L = 16 bytes											
16	benign	off	50:50	10/10	0.62	0.70	1.00	1.00	1	3
16	benign	on	R at 1%	5/5	0.50	0.33	0.04	0.03	70	56
16	benign	on	T at 1%	5/5	0.58	0.88	0.01	0.04	46	155
16	benign	on	50:50	10/10	0.70	0.52	0.05	0.05	156	134
16	lethal	off	50:50	10/10	0.63	0.32	1.00	1.00	2	2
16	lethal	on	R at 1%	5/5	0.81	0.87	0.72	0.73	3305	3476
16	lethal	on	T at 1%	5/5	0.81	0.81	0.74	0.00	3307	0
16	lethal	on	50:50	10/10	0.86	0.86	0.72	0.72	3234	3336
L = 32 bytes										
32	benign	off	50:50	10/10	0.28	0.25	1.00	1.00	1	2
32	benign	on	R at 1%	5/5	0.09	0.07	0.88	0.80	448	391
32	benign	on	T at 1%	5/5	0.08*	0.08	0.76	0.87	162	396
32	benign	on	50:50	10/10	0.05	0.06	0.85	0.84	229	308
32	lethal	on	R at 1%	5/5	0.81	0.76	0.91	0.88	4477	3879
32	lethal	on	T at 1%	5/5	0.77	0.78	0.88	0.93	4000	4272
32	lethal	on	50:50	10/10	0.76	0.77	0.90	0.92	4046	4135

World medians; Corr = corrected core (C1); H = historical core C0. Benign zero executes; lethal zero halts.
Mutation off: 100,000; on: 300,000 updates. T share = T/(R+T); core share = 5e ed b0-bearing cells /20,000.
Tail types = distinct byte 4–(L−1) sequences/world. *L32 benign T-at-1%: T median n = 4/5 defined ratios; core/tail medians n = 5/5, including zeros.

f **EXPLORATORY continuation and host controls; registered stability verdict**

Source soup	Execution zero rule	Worlds	Mean T fraction, start	Mean T fraction, end
benign	benign	10	0.071	0.070
benign	lethal	10	0.071	0.110
lethal	benign	10	0.734	0.685
lethal	lethal	10	0.733	0.740

Continuation: historical core C0 Z80 L32; end = 2,000 updates; T fraction = T/(R+T).

Host damage control: historical core C0 Z80 L32, lethal chemistry

Fraction leaving all host bytes unchanged: unmodified 0.0137; detox 0.0128.
25,600 encounters/condition (400 hosts ×64 executors).

Detox replaces payload 0x00/0x76 with non-toxic bytes; host B faces executors from its own soup.

Registered E1 not met: L16: 4/10, L32: 4/10; requires ≥7/10 worlds with final T/(R+T) in [0.3,0.7].

E1: corrected Z80 core (C1), benign chemistry, mutation off, 50:50 starts; L16/L32.
Variable endpoints do not establish neutral drift.

Extended Data 5. Environmental competitions a–d, Historical core C0 and corrected Z80 core (C1) L32 competitions under benign and lethal zero-opcode chemistry with mutation on, through 300,000 updates. Thin curves show individual worlds; thick curves show within-start-group medians. Each environment has ten 50:50 worlds and five worlds per reciprocal 1% invasion. R and T are partial-copy and full-tape byte classes. e, Complete endpoint medians for 110 worlds per implementation, covering both tape lengths, chemistries, mutation settings and starts. Mutation-off endpoints are at 100,000 updates. T share is

T/(R+T); core share counts 5e ed b0-bearing cells over 20,000 cells; tail types count distinct sequences at bytes four through L−1 per world. The corrected-core L32 benign T-at-1% median uses four defined ratios among five worlds, while core and tail medians include all five, including zeros. f, Exploratory historical-core L32 continuations report mean T share at start and 2,000 updates. The lethal L32 host control reports unchanged-host-byte fractions over 400 hosts times 64 soup-executor encounters per treatment. Detoxification replaces payload 0x00 and 0x76 with non-toxic bytes. Registered E1 under the corrected core, benign chemistry and mutation off meets its final-share interval in only four of ten worlds at each length, below seven required. Variable endpoints do not establish neutral drift.

a **6502 finite census: all completed arms**

L/BRK/budget	Periods k	Words tested	Exact self-writers
16/fait/128	1,2,3,4	4,311,810,304	0
16/fait/512	1,2,3,4	4,311,810,304	0
16/halt/128	1,2,3,4	4,311,810,304	0
16/halt/512	1,2,3,4	4,311,810,304	0
20/fait/128	1,2,3	16,843,008	0
20/fait/512	1,2,3	16,843,008	0
20/halt/128	1,2,3	16,843,008	0
20/halt/512	1,2,3	16,843,008	0
32/fait/128	1,2,3	16,843,008	0
32/fait/512	1,2,3	16,843,008	0
32/halt/128	1,2,3	16,843,008	0
32/halt/512	1,2,3	16,843,008	0

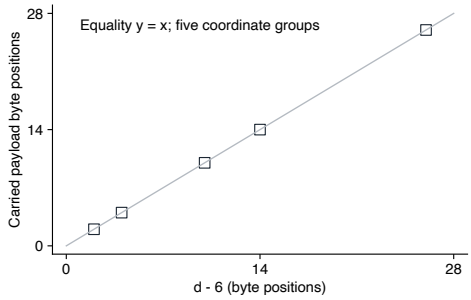
L = tape bytes; budget = execution steps; k = tiled-word period
fait = faithful BRK; halt = halt before BRK; exact output into zero B
Words tested sums k; each L16 k4 arm exhausts 256 to the fourth power words

b **8086 finite census: declared reset**

L/budget/k	Words tested	Exact output zero B	≥75% match in ≥8/16
16/128/1	256	0	0
16/128/2	65,536	20	29
16/512/1	256	0	0
16/512/2	65,536	29	29
32/128/1	256	0	0
32/128/2	65,536	26	36
32/512/1	256	0	0
32/512/2	65,536	26	25

L bytes; budget execution steps; k = word period; SP at ring end
Counts of nonzero words: exact output into all-zero byte partner B or ≥75% parent-byte match in ≥8/16 uniform-partner encounters

c **Constructed 6502: sampled carriage**



296 constructed copier/contract configurations; points coincide
d – 6 / configurations: 2/96; 4/40; 10/96; 14/32; 26/32
d = copy displacement in bytes
Carried: same-position parent byte in all 3 bodies×64 partner tests

d **S1: complete four-arm route ledger**

Arm L/budget	Worlds	Conf.	Open	Undet.	Loop (overlaps)
A 16/128	20	20	0	0	14
B 32/128	20	4	0	16	4
C 16/512	20	19	1	0	15
D 16/128	20	18	0	2	17

S1: 6502; Conf. = no partner fetch in 16/16 traces; Open = fetch ≥8/16
Undet. = no heritable copier detected; Loop = repeated instruction address
Loop overlaps routes; L bytes/budget steps; A–C faithful, D pre-BRK

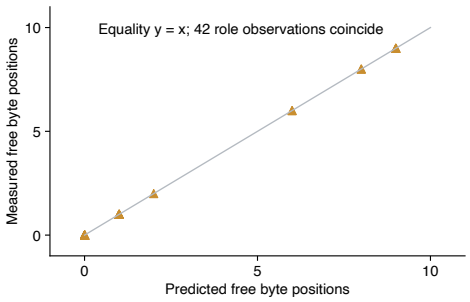
e **Registered C2: complete sampling roles**

Sampling role	Sampled	Eligible	Elig. worlds
Final random cell	640	23	4
Final top-3 by abundance	240	10	4
First detected tape	62	9	9

Eligible: 42 roles; 41 unique tapes; 11 worlds
6502 retention experiment C2: 80 S1 worlds; fixed-core certificate
Pointer randomization: unmeasured; not an 8086 S2 arm

Eligible = certified fixed-core, pure forward-copy family
942 roles; world counts overlap across roles (do not sum)

f **C2 exact capacities: all eligible roles**



Free byte positions: arbitrary alternatives jointly inherited with causal roots
Each exact-byte/core/ancestry check: 0/91,428 direct-case mismatches and 0/731,424 serial-record mismatches; 42 roles /41 tapes /11 worlds

Extended Data 6. Machine laws and finite censuses a, Complete finite 6502 census accounting by tape length, BRK contract, execution-step budget and tiled-word period, including the four-billion-word period-four arms. Exact self-writers produce exact output into an all-zero partner. b, Complete finite 8086 census counts under the declared reset; nonzero words qualify by exact output into zero partners or at least 75% parent-byte match in at least eight of 16 uniform-partner encounters. c, The 296 constructed copier/contract configurations coincide at five coordinate groups. A payload position is carried when its parent byte remains at the same offspring position for all three sampled bodies and 64 partners per body; this is distinct from a mutant-value site score. d, S1 first-route counts for four 20-world 6502 arms; open and confined are fetch classifications, whereas loops overlap routes. e, The complete 6502 retention experiment C2 sampling ledger contains 942 roles from 80 S1 worlds. Its 42 eligible roles represent 41 tapes from 11 worlds; role-level world counts overlap. f, Predicted and measured free byte positions agree for all eligible roles. Exact-byte, fixed-core and ancestry checks each have zero mismatches over 91,428 direct cases and 731,424 serial records. Free positions admit arbitrary alternatives jointly inherited with causal roots. Pointer randomization is unmeasured and was not an 8086 S2 arm.

a BFF contracts: detection and persistence

Variant	Worlds N	Modal worlds, n	Population worlds, n	Collapse n / pop. n
Standard	24	9	7	0/7
Literal, no wrap	12	12	5	5/5
Standard, harmless	12	7	3	0/3
Pointer wrap	24	19	19	0/19
Wrap + literal	12	12	12	12/12
Wrap + literal, harmless	12	12	12	0/12

Harmless: unmatched brackets do not halt; literal emits operands
BFF = Brainfuck-family machine; wrap permits pointer wrap

Modal: generation-2 parent-byte-match gain ≥0.3 (headroom-normalized)
Population: ≥50% cells heritable (same gain ≥0.3 criterion)
Later collapse: terminal heritable-cell fraction <10% after population detection

b First-modal tapes: fetch / inflow counts

Fetch/256	H (bits)	Tapes, n
Standard (n=9)		
0.00000	0.00000	4
0.00000	7.24969	1
0.00391	0.03687	2
0.00391	2.06372	1
0.19531	7.39789	1
Literal, no wrap (n=12)		
1.00000	7.97656	1
1.00000	7.98143	2
1.00000	7.98438	2
1.00000	7.99219	2
1.00000	8.00000	5
Wrap + literal, harmless (n=12)		
1.00000	0.00000	4
1.00000	0.03687	6
1.00000	0.07373	1
1.00000	0.11056	1

Fetch/256	H (bits)	Tapes, n
Pointer wrap (n=19)		
0.00000	0.00000	11
0.00000	0.03687	2
0.00000	1.92163	1
0.00000	2.52233	1
0.00000	7.07753	1
0.00000	7.16823	1
0.00781	0.07373	1
0.00781	1.73558	1
Wrap + literal (n=12)		
1.00000	0.58729	3
1.00000	0.62380	1
1.00000	0.69675	1
1.00000	0.73319	1
1.00000	0.76960	2
1.00000	0.80599	2
1.00000	0.84236	1
1.00000	0.98759	1

64 first-modal tapes, one/world; n counts each fetch/H pair
Fetch: encounters with any partner fetch /256; values rounded to 5 decimals
H = H(offspring | fixed parent), bits; 256 uniform partners
Five variants shown; Standard, harmless has no entropy archive

c Open fetch; zero inflow for r ≥ 2

Operand copies r	Worlds	H bits	Fetch/256
1	12	8.0	1.00
2	6	0.0	1.00
3	6	0.0	1.00

Literal, no wrap; r = 1 baseline; r counts operand copies
Literal P a b emits a,b repeated r times

World medians: conditional H and any-partner-fetch encounters/256
r≥2: zero partner inflow with fetch in every encounter

d Halting dial: all worlds included in counts

Halt p	Worlds N	P persistence worlds, n/N	Median final P share all worlds
0	12	12/12	0.984
0.01	6	6/6	0.984
0.03	6	6/6	0.984
0.1	6	6/6	0.984
0.3	6	5/6	0.102
1	12	0/12	0.002

Background: Wrap + literal; p = 0 harmless, p = 1 halting

p = probability an unmatched bracket halts; P = literal-copy instruction
Persistence: P-tape share ≥10% at 16,384 BFF population updates (epochs)

e First / final inflow and fetch

Variant	Worlds n	H bits first / final	Fetch/256 first / final
Literal, no wrap	12	7.99 / 7.99	1.00 / 1.00
Standard	9	0.04 / 0.00	0.00 / 0.00
Pointer wrap	19	0.00 / 0.00	0.00 / 0.00
Wrap + literal	12	0.75 / 0.64	1.00 / 1.00
Wrap + literal, harmless	12	0.04 / 0.00	1.00 / 1.00

First = first-modal tape; final = terminal modal tape

Five variants shown; Standard, harmless has no entropy archive
Paired world medians; fetch = any-partner-fetch encounters /256

f Registered tests of partner inflow (H)

Per-world criterion H in bits	Passing worlds, n/N	Support / reject % passing	Verdict
Q1 Z80 first H>1	80/80	≥90% / ≤60%	met
Q2 Z80 loop final H<0.5 67 worlds: loop final tapes	63/67	≥90% / ≤60%	met
Q3 Z80 H_first – H_final ≥2	63/80	≥75% / ≤50%	met
Q4a BFF standard first H<0.5	6/9	≥90% / ≤60%	between
Q4b BFF pooled literal variants: first H>1	12/36	≥90% / ≤60%	rejected

Q4b: Literal, no wrap; Wrap + literal; Wrap + literal, harmless
12 worlds each; 36 total; same names as a, b and e

Q1–3: Z80 Stage G (80 worlds); Q2: 67 worlds with loop-bearing final tapes
Q4a: Standard BFF; Q4b: the three pooled variants named above
Support/reject use pass/world fraction; between = neither threshold

Extended Data 7. BFF literal and environmental controls a, All archived BFF variants with modal detection, population detection and later collapse counts. Modal detection requires headroom-normalized generation-two parent-byte-match gain of at least 0.3; population detection requires at least half of sampled cells to satisfy the same heritability criterion. Collapse is terminal heritable share below 10% after population detection. Literal writing emits operands; wrap permits pointer wrap; harmless variants prevent unmatched brackets from halting. b, Complete paired fetch-fraction and conditional-entropy counts for first-modal tapes, one per detected world. All distinct observed pairs have explicit tape counts; values are rounded to five decimals. H is offspring entropy conditional on the fixed parent, measuring partner inflow over 256 uniform partners per tape. The standard harmless-bracket variant has no archived entropy panel. c, Literal-no-wrap repetition controls: open fetch can coexist with zero sampled partner inflow. d, Wrap-plus-literal halting controls retain all worlds in counts and final P-tape-share medians at 16,384 BFF population updates, called epochs in the archive. e, Paired first-versus-final world medians of entropy and fetch. f, Registered individuality results pair each pass fraction with its support and rejection thresholds.

Z80 claims use 80 Stage G worlds, except the 67 loop-bearing finals; BFF Q4b pools three literal variants of 12 worlds each. Q4b is rejected and Q4a lies between decision boundaries. Entropy, confinement and dependence on initial partner contents remain separate measurements.

a **Genealogy cohorts: worlds and nested lines**

Cohort	Worlds	Confined worlds	Lines from confined worlds	Open-first lines
N3-v6	20	17	1088	1088
N6	20	20	1280	1280

64 sampled-cell ancestries / confined world; world = simulation
N3-v6: discovery; N6: confirmation; Z80 tape length L = 16 bytes
Copier: ≥75% tape bytes copied in ≥32/64 uniform partners
Counted trace = fetch of any partner opcode or operand byte
Genealogy fetch class: open ≥32/64 counted traces; confined <32/64
Open-first: earliest copier on each sampled line is classified open

Sampled lines: 64 ancestries per world with a confined founder
Worlds column counts all worlds; genealogy classes use 64 partner traces

b **First founders versus later conversions**

Cohort	First founders, worlds	Later founder events, n	Total founder events
N3-v6	17	15	32
N6	20	14	34

20 worlds/cohort; at most one first confined founder/world
Founder: first confined copier after latest open copier on sampled line
Open ancestry to confined copier; classification as in panel a
Later events follow the first founder in that world; events can repeat
Later events can reuse existing confined-cell byte sources

First founders and later confined-founder events stay separate
Worlds, ancestral lines and founding events are different units

c **N6 first-founder mechanisms**

Recorded mechanism	Founders /20
novel	18/20
damage	2/20

novel = rewrite, <75% byte match to either parent (18)
damage = partial overwrite, ≥75% parent byte match (2)

Separate N6-1 source rule: copy/rewrite 18/20, mutation 0/20
Recorded event categories and causal source classes differ

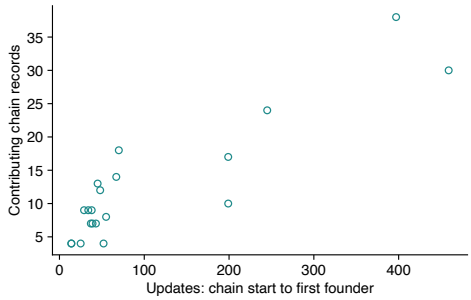
d **N6 founding executors: two copier scopes**

Executors, n	Whole-tape	Executed-byte
Non-copier	20	11
Copier	0	9

Executed-byte: 11/20 non-copiers; requires ≥12/20; not met
Whole-tape: 20/20 non-copiers; requires ≥16/20
Soup-abundance null: P(all 20 whole-tape non-copiers) = 0.062
Null draws executors by pre-founder soup composition

One founding executor/first confined founder: 20 N6 worlds
Copier: ≥75% scoped bytes copied in ≥50% partner encounters
Scopes: entire tape / only executed instruction-stream bytes

e **N6: all first-founder chain lengths**



N6: one point per first founder, 20 worlds
Records = contributing chain events; intervening whole-tape donors are non-copiers (executed-byte function kept separate)

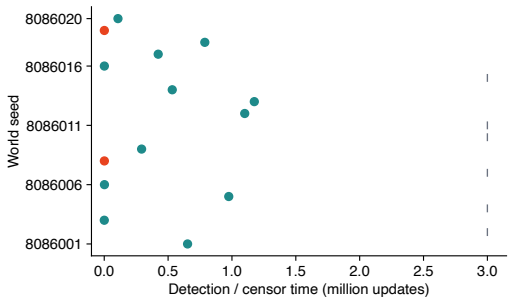
f **Registered N6: complete component ledger**

ID	Requirement	Observation	Verdict
N6-1	Copy/rewrite ≥16/20; mutation ≤2/20 founders	Copy/rewrite 18/20 Mutation 0/20	met
N6-2	Whole-tape non-copier chains ≥15/20	20/20	met
N6-3	Median cell carrier share ≥5× random-tape baseline	Observed 0.00190 Baseline 0.00024 Ratio 7.9×	met
N6-4	Non-copier executors whole-tape ≥16/20; executed-byte ≥12/20	Whole 20/20 Executed-byte 11/20 Null P = 0.062	not met
N6-5′	Expected mutation founders <0.05 in 20-world cohort	0.0064 expected 0/20 observed	met

N6-3: cells carrying a word / all soup cells; 2-byte word at executed start
Median across 99 word/founder rows in pre-founder snapshots
Mutation-founded first founders: 0/20; N6 rejection threshold ≥5/20
Input: 55 point-mutation events on 20 pre-founder chains

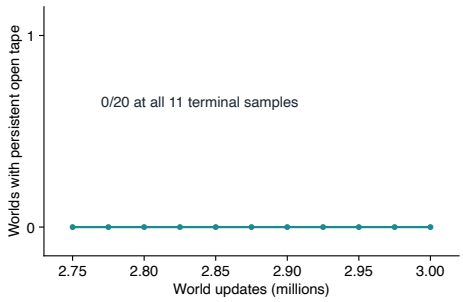
Extended Data 8. Founder genealogies a, Z80 L16 discovery and N6 confirmation each comprise 20 worlds. Sixty-four ancestral lines are sampled from each world with confined founders: 17 discovery and 20 N6 worlds. The genealogy classifier requires at least 75% tape-byte copying in at least 32/64 uniform partners; its open/confined classes count any partner opcode or operand fetch in at least/fewer than 32/64 encounters. Open-first refers to the earliest copier on each line. b, First founders and later open-ancestry-to-confined-founder events have separate denominators. c, N6 founding-event categories distinguish rewrites from partial overwrites; recorded mechanisms and registered source classes are separate. d, One executor per N6 first founder is classified under whole-tape and executed-byte copying scopes. A copier reproduces at least 75% of scoped bytes in at least half of partner encounters. All 20 executors are whole-tape non-copiers, but only 11 are executed-byte non-copiers, below 12 required. The probability of all 20 being whole-tape non-copiers under independent draws from pre-founder soup composition is 0.062. e, Chain records and elapsed updates for all 20 first founders. f, Every registered N6 component includes its decision boundary and observation. N6-3 measures cells carrying two-byte words at founder-executed instruction starts divided by all soup cells; its median over 99 word/founder records is 0.00190 against a 0.00024 random-tape baseline, a 7.9-fold ratio. The 55 recorded point-mutation events on 20 pre-founder chains imply 0.0064 expected confined founders; zero of 20 founders are observed mutation completions. N6-4 remains not met; the five-of-20 point-mutation kill is not triggered.

a **8086 S2: registered first-copier detection**



S2: coherent-fetch emergence; 20 worlds; L = 16 bytes
14/20 detected; 6/20 censored at 3 million updates
Teal dots: confined, 0/16 traces; red dots: open, ≥8/16 traces
Counted trace = any partner opcode/operand byte fetch
No first cases at 1–7/16; grey ticks = right censor

b **R2: descriptive open-copier persistence**



Same open fetch class as a; persistence also requires memory feeding
Same 20 S2 worlds; terminal persistence = 0/20
All earlier scheduled samples combined: 1/20 worlds ever positive

c **Registered evolved law: untested**

Family test	Candidate tapes	Eligible tapes	Unresolved tapes
L1	1337	0	0
L2	1337	0	724
L2 excluding L1	1337	0	724

L2 support suspended: zero certified cases

L1: frozen overlap; L2: broader pure-copy certificate

Zero eligible cases: law tests untested; capacity undetermined

c–e: candidate tapes from same 20 S2 worlds as a
L1 frozen overlap / L2 broader pure-copy certificate; d = L2 ledger

d **Registered exclusions and unresolved cases**

Status	Reason	Candidate tapes, n
culture-negative	culture not positive	558
excluded	original partner read	55
unresolved	Opcode 07 (POP ES)	708
unresolved	Opcode 0C	2
unresolved	Opcode 0F	1
unresolved	Opcode 21	1
unresolved	Opcode 31	1
unresolved	Opcode 5F	10
unresolved	Opcode E5	1

Total: 1,337 candidate tapes; 724 unresolved

L2 candidate tapes: 558 culture-negative, 55 excluded
724 unresolved (708 at POP ES); total 1,337

e **Post hoc: admissions and exact-law checks**

d bytes / edit	Genotypes	Worlds	Encounters with controls	Byte/core/ancestry mismatches
4 / erasure	10	10	538,240	0
16 / single/joint payload	22	20	1,686,400	0

d = write-minus-read displacement mod 32, in bytes
Paired address ring: donor A 0–15, partner B 16–31

Same 20 S2 worlds: 779 culture-positive world/tape records

Then 600 admitted records (590 d = 16; 10 d = 4); 32 selected genotypes

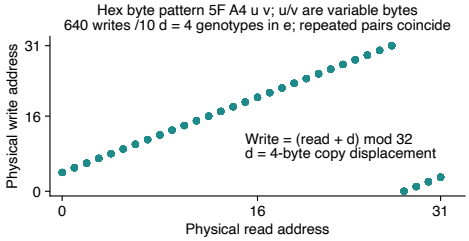
d = 4 joint free-payload tests: 0 applicable positions/genotype (n = 10)

d = 16 erasure tests: 0 applicable positions/genotype (n = 22)

32 genotypes /20 worlds; 2,224,640 encounters; worlds overlap by d

Observed offspring bytes, fixed core and ancestry match copy-law predictions
Nonempty perturbations and empty-assignment controls included in table
No applicable coordinates above means no corresponding perturbations
Listed encounters include empty-assignment controls; check bytes/core/ancestry

f **Exploratory byte-pattern copies: d = 4 bytes**



Separate snapshot fates: pooled median targets /512 encounters

Cohort	Median targets
8086 S2: 20 L16 worlds; 3 donor motifs	94.5
Z80 Stage G: 20 L16 worlds; 1 donor motif	214.5

Target: new non-donor 16-byte half matching a donor 2-byte tiling

Pools 5 snapshots/world × both soup and uniform partners × donor A/B
G = random-code cohort; motifs = distinct tiled 2-byte donor sequences

Extended Data 9. Registered S2 and separate follow-ups a, All 20 registered coherent-fetch 8086 L16 S2 worlds: 14 first-copier detections and six right-censored nondetections at 3,000,000 updates, with open and confined fetch labels. b, Descriptive R2 persistence in the same worlds: zero open memory-fed tapes at all 11 terminal samples; earlier scheduled samples combined contain one ever-positive world. c–d, Registered law accounting from these S2 worlds retains culture-negative, excluded and unresolved candidate tapes. No case is certified eligible: L1/L2 remain untested and L2 support is suspended. e, Separate exploratory admissions and selected post hoc exact-law assays from the same 20 worlds: 779 culture-positive world/tape records lead to 600 admissions, comprising 590 d16 and ten d4 tapes, and then 32 selected genotypes. All 2,224,640 encounters check predicted bytes, core and ancestry without mismatch. Copy displacement d is the write-minus-read byte offset. Nonempty perturbations comprise d4 erasure and d16 single/joint payload; d4 joint payload and d16 erasure have zero applicable coordinates and no corresponding perturbations. Empty-assignment controls remain in encounter counts. f, Exploratory geometry for ten d4 genotypes with

hexadecimal byte pattern 5F A4 u v, where u/v are variable bytes; repeated read/write pairs coincide. The separate post hoc snapshot-fate result pools five samples per world, soup/uniform partners and donor A/B roles; medians are totals per 512-encounter batch of non-donor 16-byte halves newly matching the donor's two-byte tiling, across 20 S2 and 20 Z80 G L16 worlds. S2 uses three donor tiles and Z80 one. These follow-ups do not supply registered evolved-law evidence.